Pact

Developer & Integrator Handbook

First Edition — June 2026

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Copyright and disclaimer

> **Disclaimer** — Pact is the swap engine that powers trustless, peer-to-peer atomic swaps. It holds keys and signs and broadcasts transactions on live networks. This handbook documents the engine for developers, integrators, and operators; it is **not** financial advice and makes no guarantees. The software is provided "as is", without warranty of any kind. The code is the ultimate source of truth; consult the official project repository for the most up-to-date information.

Part I

Overview

Chapter 1

About this Handbook

This is the **Pact Developer & Integrator Handbook** — the reference for the software that actually executes atomic swaps in the Satchel project. It is written for people who run, drive, or build against the swap engine, not for end users trading from the desktop app. If you are an end user, read the *Satchel User Handbook* instead; this handbook assumes you are comfortable with Bitcoin script, key derivation, JSON-RPC, and the Rust toolchain.

1.1 Who this is for

You will get the most out of this handbook if you are one of the following:

- An **operator** running `pactd` — the local swap daemon — as a long-lived service: wiring it to your Bitcoin-Core-compatible nodes, configuring coins and confirmation depths, managing seeds and merchants, and keeping the auto-refund scheduler alive.
- A **front-end or tooling developer** building against the engine's JSON-RPC 2.0 API: a custom UI, a bot, a monitoring dashboard, or an integration that posts and takes offers programmatically.
- A **protocol implementer** who wants to understand — or independently reimplement — the on-chain swap construction: the HTLC scripts, the Taproot/MuSig2 adaptor scheme, the key-derivation paths, the timelock margins, and the message flow.

  Note — The canonical, machine-checkable description of the wire format and on-chain construction lives in the `spec/` directory of the repository, together with deterministic test vectors. This handbook explains and contextualises that material; where the two ever disagree, the spec and the code win.

1.2 How this handbook is organised

The handbook is grouped into six parts that move from orientation to deep detail:

1. **Overview** — what Pact is, the crate map, and the architecture and trust boundaries. Start here.

2. **Running Pact** — building from source, running `pactd`, configuring coins and pairs, and managing seeds, wallets, and merchants.
3. **The JSON-RPC API** — the complete method surface: node/info, the seed/wallet lifecycle, coins and pairs, v1 and v2 swaps, board operations, private offers, and fees.
4. **The Swap Protocol** — the v1 HTLC and v2 adaptor state machines step by step, the on-chain scripts, timelocks and action deadlines, fee management, and auto-refund.
5. **Transports** — how identity-signed offers and sealed coordination blobs move over Nostr relays and the self-hostable Corkboard.
6. **Building Against Pact** — driving swaps with `pact-cli`, the generic `call` escape hatch, and end-to-end worked examples.

You do not have to read straight through. Operators can jump to the chapter *Running pactd*; integrators to *The pact-cli* and the API part; implementers to the protocol part.

1.3 Conventions

This handbook follows a few consistent conventions:

- **Callouts** highlight things worth pausing on:

  Tip — a shortcut or a recommended way to do something.

  Note — context, a clarification, or a cross-reference.

  Warning — something that can lose funds, corrupt state, or fail boot if you get it wrong.

- **Text styles** carry meaning. `monospace` marks commands, flags, RPC method names, field names, file paths, and any exact text you type. **Bold** marks component names and UI labels. *Italics* mark a new term on first use.
- **Cross-references** name the target chapter by its title — for example, *see the chapter "Running pactd"* — never by number, because numbers shift as the handbook grows.
- **Code blocks** are fenced with a language hint (`sh`, `json`, `rust`, `text`) and are meant to be copy-pasteable.

1.4 Version and status

Status — Both swap protocols — v1 (HTLC) and v2 (Taproot/MuSig2 adaptor) — are reviewed, audited, and live on mainnet.

The handbook is kept in sync with the `master` branch, and the **code is the ultimate source of truth**. When a precise detail matters for your integration, confirm it against the code and the pinned test vectors before you rely on it. Where this handbook cites behaviour, it reflects the engine as of the `master` branch it was written against.

1.5 Where to get help

- The repository `README.md` and the `docs/` directory carry the architecture and per-feature design notes.
- The `spec/` directory is the authoritative protocol description plus the `htlc_v1.json` / `htlc_v2.json` test vectors.
- The end-to-end harness under `pact/harness/` (notably `test_swap_e2e.py` and `test_adaptor_swap.py`) is the best executable reference for how a full swap is driven — when in doubt, read the harness.

Chapter 2

What Pact Is

Pact is the swap engine at the heart of the Satchel project: the software that builds, signs, broadcasts, and watches the on-chain transactions that make a trustless cross-chain trade *atomic*. It is the component that holds your keys, enforces the timelocks, and refunds you if a counterparty walks away. Everything else in the project — the desktop app, the transports — exists to feed offers to Pact and render what Pact reports back.

An *atomic swap* is a trade between two chains in which either both legs settle or neither does. Pact implements this two ways: *v1*, a classic hash-locked HTLC swap, and *v2*, a Taproot/MuSig2 *adaptor* swap. Both are covered in detail in the protocol part of this handbook.

2.1 The crate map

Pact is a small Rust workspace plus two supporting crates. Knowing which crate owns what is the fastest way to navigate the codebase:

Crate	Role
`libswap`	The engine proper: the HTLC and adaptor logic, the on-chain script construction, the key derivation, and the two swap **state machines**. All swap intelligence lives here; everything else is a shell around it.
`pactd`	The local daemon. Exposes `libswap` over **JSON-RPC 2.0**, persists state in **SQLite**, and runs the background scheduler that auto-refunds, auto-redeems, and RBF-fee-bumps with no human present.
`pact-cli`	A thin RPC client — a hand-rolled HTTP caller that maps subcommands (and a generic `call`) onto `pactd` methods. The integrator's command line.

Crate	Role
`pact-proto`	The wire format and crypto primitives: signed envelopes, canonical JSON, BIP340 signatures, recipient-sealed relay blobs (`PACTSEALED1`), and private-offer slips.
`pact-nostr`	Pure mapping between Pact envelopes and Nostr events (public offer adverts as kind `31510`, gift-wrapped relay messages as kind `1059`). Mapping and crypto only — no relay I/O.

> **Note** — `libswap` is where you look to verify how a swap is constructed; `pactd` is where you look to verify how it is *driven* and *persisted*. The two state machines (`swap::State` for v1, `AdaptorState` for v2) are defined in `libswap` and stepped by the engine's scheduler tick inside `pactd`.

2.2 The bitcoind analogy

If you have run Bitcoin Core, the shape of Pact will be familiar — it is deliberately the same:

- `pactd` ≈ `bitcoind` — a headless daemon you run and leave running, with a data directory, a `.cookie` file for RPC auth, and a JSON-RPC endpoint.
- `pact-cli` ≈ `bitcoin-cli` — a thin command-line client that reads the cookie and talks to the daemon.
- **Satchel** ≈ `bitcoin-qt` — the optional desktop GUI that bundles and supervises the daemon and renders its RPC. (Satchel is documented in its own handbook.)

The data layout, the cookie auth, the HTTP Basic credentials, and the loopback-only listener are all modelled on Core. If you know how to run and script `bitcoind`, you already know most of how to run and script `pactd`.

2.3 What Pact deliberately is *not*

Pact is a swap *executor*, and nothing more. It is important to be precise about the boundaries:

- **No custody.** Pact never holds your coins on your behalf in any account. Your funds are either in your own node's wallet or locked in a swap output that only you or your counterparty can spend — enforced by the chain, not by Pact.
- **No matching engine.** Pact does not pair buyers with sellers. It posts an offer to a transport and takes an offer from one; *finding* the offer is the transport's job (see the chapter *Architecture & Trust Boundaries*).
- **No fees.** There is no platform fee anywhere. The `platform_fee_sat` field reported by `estimateswapfees` is hard-wired to `0`; the only costs in a swap are the on-chain miner fees of its transactions.

- **No order book of record.** Offers live on transports (Nostr relays or a Corkboard), which are untrusted and replaceable. Pact derives which trading *pairs* are possible from coin capabilities; it does not curate a market.

2.4 The market metaphor

The project's naming follows a village-market-square theme, and it maps cleanly onto the components: a *pact* is the trustless deal itself, *posted on the corkboard* (the transport), and *settled into your satchel* (the app and its balances). There is deliberately no "exchange" or "DEX" branding — Pact is a protocol and a daemon, not a venue.

2.5 The first pair

The first supported trading pair is **BTCX** ↔ **BTC**, where *BTCX* is Bitcoin-PoCX. More coins follow — Litecoin (LTC) was the first added third coin — and new Bitcoin-Core-compatible chains can be added as data without recompiling the engine (see the chapter *Coins, Pairs & Capabilities*). For the shipped BTCX ↔ BTC pair the engine selects the v1 HTLC protocol by default; the v2 adaptor protocol is used for Taproot pairs that lack an HTLC option.

Chapter 3

Architecture & Trust Boundaries

Pact's architecture is built around one rule: **the things that can lose your money never leave your machine, and the things that leave your machine can never lose your money.** Understanding where that wall sits is the single most important thing for anyone integrating against, operating, or auditing the system.

3.1 The three moving parts

A running swap involves three categories of component, with a hard trust wall between the first two (yours) and the third (hosted and untrusted):

1. **Your machine — fully trusted.** This is the stack you run and control:
    - **Satchel** (or any RPC client) — the face. It renders the engine's RPC and never touches swap logic.
    - **`pactd`** — the swap engine. It owns the BIP39 *seed*, derives every key, builds and broadcasts the swap transactions, and runs the scheduler that auto-refunds if a counterparty disappears.
    - Your **coin backends** — a BTCX node and a BTC node (or Electrum backend), reached over their own RPC. These hold the actual coins and the funding wallet.
2. **Hosted, untrusted transport.** A **Nostr relay** (the default) or a **Corkboard** instance (self-hostable). Its only job is to carry identity-signed offers and forward sealed coordination blobs between counterparties. It is replaceable, runs no swap logic, and is assumed hostile.

    Note — "Three moving parts" counts the transport as one part even though you can point Pact at several relays or boards at once. They are interchangeable couriers, not authorities.

3.2 The hard wall

The boundary is precise and worth stating in operational terms:

- **Keys and refunds are local, always.** The seed, every derived key, the preimage (v1) and adaptor secret (v2), the pre-signed refund transactions, and the entire decision to redeem or refund live inside `pactd` on your machine. None of this is ever sent to a transport.
- **The transport sees only signed offers and sealed blobs.** What leaves your machine is (a) an *identity-signed offer* — an advert that you are willing to trade, signed by your BIP340 identity key so it cannot be forged or altered — and (b) *recipient-sealed coordination blobs* (`PACTSEALED1`), encrypted to the counterparty so the relay operator sees ciphertext only. A relay cannot read your coordination, cannot move your funds, and cannot match, execute, custody, or charge.
- **The RPC is loopback-only.** `pactd` binds `127.0.0.1` by default and **enforces** that its listen address is a loopback address — a non-loopback `--listen` aborts boot. The JSON-RPC surface is for local clients (the CLI, Satchel) reading the on-disk cookie; it is never meant to face a network.

  Warning — Because the RPC has no remote-access design, do not put `pactd` behind a reverse proxy or expose its port. Anything that can reach the RPC and read the `.cookie` can drive your swaps and your wallet. Keep it loopback.

3.3 Data flow of an offer

At a high level, posting and discovering an offer looks like this:

1. You call an offer-posting RPC on `pactd` with the coins, amounts, and timelocks.
2. `pactd` builds an offer envelope and **signs it with your identity key** (the BIP340 key at `m/7228'/0'/0'`, which never appears in any on-chain script).
3. `pactd` hands the signed envelope to each configured transport, which publishes it (Nostr kind `31510`, or a Corkboard row).
4. A counterparty's `pactd`, browsing that transport, sees the signed advert, verifies the signature, and — if they want it — *takes* it, which kicks off the coordination flow.

The transport stored and forwarded a signed advert. It never learned a key.

3.4 Data flow of a swap

Once an offer is taken, the two daemons coordinate to build and settle the swap:

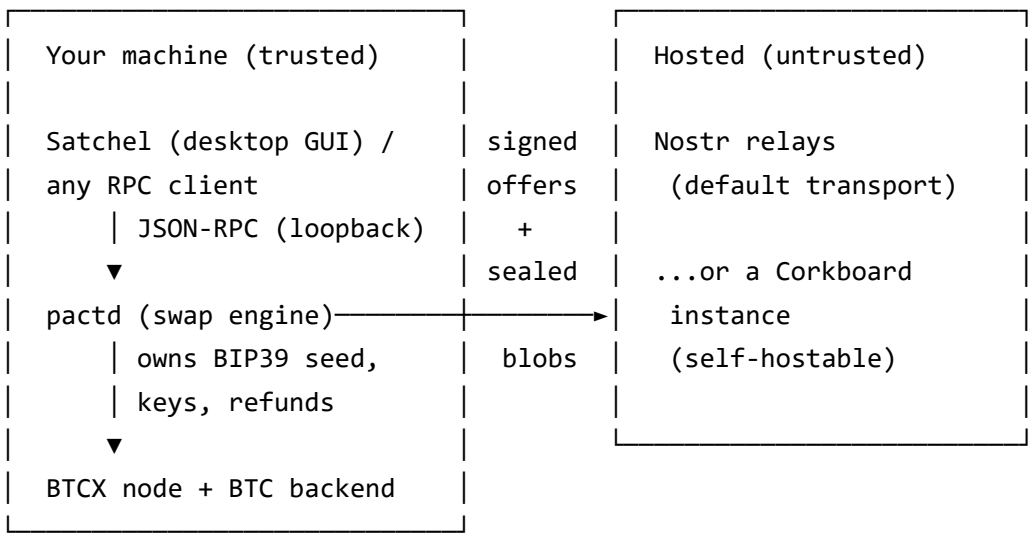
1. The takers exchange a short sequence of **sealed messages** through the transport — funding details, public keys, sweep addresses, and (for v2) the MuSig2 nonces and adaptor signatures. Each message is sealed to its recipient (`PACTSEALED1`), so the relay forwards opaque ciphertext.
2. Each side independently **reconstructs the on-chain output** (the HTLC P2WSH for v1, or the Taproot output for v2) and byte-compares it before any funds move — neither side trusts the other's claim about the script.
3. The maker funds first; the taker verifies the funding on-chain, then funds its leg. Redemption on one chain reveals the secret (the preimage for v1, the adaptor secret `t` for v2) that

lets the counterparty claim the other chain.

4. Throughout, each `pactd` scheduler tick watches both chains and, if a deadline passes with the swap unfinished, broadcasts the pre-signed (v1) or re-derived (v2) **refund** — no human and no transport required.

The detailed message sequence, the exact scripts, and the timelock margins are covered in the protocol part of this handbook — *see the chapters "What Pact Is" and the swap-protocol chapters*. The point here is structural: all of step 2 through step 4 happen between two local engines and two sets of nodes; the transport only relayed sealed blobs in step 1.

3.5 The picture



Everything left of the arrow is yours and trusted. Everything right of it is hosted, replaceable, and assumed hostile. The arrow carries only signed adverts and sealed ciphertext — never a key, never custody, never a fee.

Part II

Running Pact

Chapter 4

Building from Source

Pact is a Rust workspace. Building the engine and running its tests needs only the Rust toolchain; running the full end-to-end harness additionally needs Python 3 and a regtest `bitcoin-pocx` build. This chapter covers building the engine, running the test suites, building the Corkboard transport, and where the binaries land.

4.1 Prerequisites

- **Rust** (stable) with `cargo`. The whole engine — `libswap`, `pactd`, and `pact-cli` — builds with a standard stable toolchain; no nightly features are required.
- **Python 3**, for the end-to-end harness scripts under `pact/harness/`.
- A **`bitcoin-pocx` build** providing the regtest BTCX and BTC nodes the harness spins up. The harness drives real `bitcoind`-shaped nodes on regtest; it does not mock the chain.

  Note — The desktop app **Satchel** adds a Node/npm + Tauri toolchain on top of this. That build is covered in the *Satchel User Handbook*; this handbook concerns the engine and its transports only.

4.2 Building and testing the engine

From the `pact/` directory:

```
cd pact
cargo build          # build libswap, pactd, pact-cli
cargo test           # unit tests + protocol-vector tests (v1 + v2)
```

`cargo test` runs the unit tests across the workspace and the deterministic **protocol-vector** tests. Those vector tests (`pact/libswap/tests/vectors.rs` for v1 and `vectors_v2.rs` for v2) pin the on-chain construction against the JSON vectors in `spec/vectors/` (`htlc_v1.json`, `htlc_v2.json`). If you change anything that touches script construction, key derivation, or the adaptor mechanism, these are the tests that must stay green.

4.3 Running the end-to-end harness

The harness performs complete swaps on regtest against real nodes — the highest-fidelity check that the engine works end to end:

```
python harness/test_swap_e2e.py      # full BTCX↔BTC v1 (HTLC) swap on regtest
python harness/test_adaptor_swap.py  # full v2 (Taproot/MuSig2 adaptor) swap
```

`test_swap_e2e.py` exercises the v1 flow: the manual two-CLI happy path, refund (including premature- and late-reveal negatives), the daemon autopilot (scheduler-driven auto-redeem and RBF fee-bump, and auto-refund on both sides), chain-watched funding, balance validation, encrypted-seed create/import, coin setup, and the Corkboard, Nostr-relay, and private-offer flows.

`test_adaptor_swap.py` exercises the v2 flow: the happy path, the single-key CLTV-tapleaf refund and its fee-bump, the CPFP redeem-bump (on BTC and on litecoind), the reveal depth-gate, and the v2 Corkboard flow.

> **Tip** — Both scripts share `regtest_harness.py`, which always brings up BTCX and BTC nodes and only adds an LTC node when constructed with `Harness(with_ltc=True)`. The harness uses `setmocktime` to advance median time on both chains, so timelock-dependent paths (refund, deadline gates) can be tested deterministically without waiting in real time.

4.4 Building Corkboard

Corkboard is the self-hostable transport — a single `axum` + SQLite/Postgres binary that stores signed offers and blind-relays sealed blobs. Build and run it from its own directory:

```
cd corkboard
cargo build
cargo run -- --listen 127.0.0.1:9780 --db corkboard.sqlite
```

Its default listen address is `127.0.0.1:9780`. Corkboard is optional: the default transport is **Nostr**, which needs no infrastructure — you simply point `pactd` at relays. See the transports part of this handbook for how to wire each one.

4.5 Where the binaries land

`cargo build` produces unoptimised binaries under the workspace `target/debug/` directory; `cargo build --release` produces optimised ones under `target/release/`. From the `pact/` workspace you get `pactd` and `pact-cli`; from `corkboard/` you get the `corkboard` binary. You can run any of them in place with `cargo run -p <crate> -- <args>` without copying the binary out — for example:

```
cargo run -p pactd -- --data-dir ./data --coin btcx=<rpc-url> --coin btc=<rpc-or-electrum-ur
cargo run -p pact-cli -- --data-dir ./data getinfo
```

The next chapter, *Running pactd*, covers the daemon's full command line, RPC authentication, and the scheduler in detail.

Chapter 5

Running pactd

`pactd` is the swap daemon — the long-lived process that exposes the `libswap` engine over JSON-RPC 2.0, persists state in SQLite, and runs the scheduler that auto-redeems, auto-refunds, and RBF-fee-bumps with no human present. This chapter is the operator's reference: the full command line, every flag, RPC authentication, and the scheduler model.

5.1 Synopsis

`pactd` is a JSON-RPC 2.0 daemon served over **HTTP POST /** (plus an unauthenticated `GET /health`), loopback-only:

```
pactd --data-dir <DIR> [--coins-file <coins.toml>] [--coin <id>=<url[,url]> ...]
      [--coin-confs <id>=<N> ...] [--listen <addr:port>] [--network <net>]
      [--board-url <url[,url]>] [--nostr-relay <wss://…[,…]>] [--auto-fund]
      [--tick-secs <s>] [--once] [--auto-init] [--merchants]
```

Only `--data-dir` is required. With no `--coin`, the daemon starts but cannot move funds on any chain until a backend is attached.

5.2 Flags

Flag	Type	Default	Meaning
`--data-dir`	path	**required**	Data directory holding the seed, SQLite state, `.cookie`, and `pact.conf`. In `--merchants` mode this is the *parent* of `merchants/<id>/`.

Flag	Type	Default	Meaning
`--coins-file`	path	none	A `coins.toml` adding coins beyond the two built-ins (`btcx`, `btc`); merged at startup. A file coin whose id collides with a built-in is dropped. A bad file logs and falls back to the built-ins rather than refusing boot.
`--coin`	repeatable `id=url[,url]`	none	Per-coin chain backend. The first URL is the wallet-qualified Core-RPC primary that funds swaps; the rest may be Electrum (`tcp://` / `ssl://`). The coin id must be in the registry; the last `--coin` for a given id wins.
`--coin-confs`	repeatable `id=N`	network/spacing default	Per-coin confirmation depth (reorg finality, `N ≥ 1`). Gates auto-redeem and completion in v1 and v2. Omitted coins use the default heuristic (see the chapter *Coins, Pairs & Capabilities*).
`--listen`	`addr:port`	`127.0.0.1:9737`	JSON-RPC listen address. **Must be loopback** — this is enforced; a non-loopback address aborts boot.
`--network`	string	`regtest`	One of `regtest`, `testnet`, `mainnet`.

Flag	Type	Default	Meaning
`--board-url`	string	none	Corkboard base URL(s), comma-separated (the HTTP transport).
`--nostr-relay`	string	none	Nostr relay `wss://…` URL(s), comma-separated. Runs **alongside** any `--board-url`; empty or absent disables it.
`--auto-fund`	flag	false	Auto-fund our HTLC leg in board-driven swaps (a griefing trade-off — only enable for unattended makers you accept that risk for).
`--tick-secs`	u64	`30`	Scheduler interval in seconds; `0` disables the background loop entirely.
`--once`	flag	false	Run a single scheduler pass (`sync_board` + `tick`), print the resulting events as JSON, and exit. Exit code is `1` if any event has `action == "error"`.
`--auto-init`	flag	false	Create the seed + state on first run (flat layout). No-op if a seed already exists.

Flag	Type	Default	Meaning
`--merchants`	flag	false	Use the nested `merchants/<id>/` layout with an in-process registry. Without it, the legacy **flat** layout (one seed in the data-dir root) is used. Ignored once a flat seed already exists.

> **Note** — The default `--network` is `regtest`. For a real deployment you must set `--network mainnet` (or `testnet`) explicitly; the network selects the chain params, the timelock-margin profile, and the confirmation-depth defaults.
>
> **Warning** — `--listen` is loopback-enforced by design. Do not try to bind a routable address to share a daemon — it will refuse to start. The RPC has no remote-access model; see the chapter *Architecture & Trust Boundaries*.

5.3 RPC authentication

Authentication is HTTP Basic, modelled on `bitcoind`:

- **Cookie (always on).** On startup `pactd` writes `<data-dir>/.cookie` containing `__cookie__:<32-byte hex>`, regenerated per run, and removes it on clean shutdown. Local clients (the CLI, Satchel) read this file — this is the zero-config default.
- **`pact.conf` (optional).** A `<data-dir>/pact.conf` file of `key = value` lines (`#` for comments) may set `rpcuser` and `rpcpassword`; those credentials are accepted *alongside* the cookie.
- Both forms are expanded to `Basic <base64(user:pass)>` and compared in constant time. The `GET /health` endpoint is unauthenticated.

> **Note** — To open an *encrypted* seed at boot without an interactive unlock, set the `PACT_PASSPHRASE` environment variable; the daemon uses it to decrypt the seed on startup. See the chapter *Seeds, Wallets & Merchants*.

The default RPC port is **9737**. (Coin peer-to-peer ports live in each coin's chain params and are unrelated to the RPC port.)

5.4 The scheduler tick model

`pactd` runs a background loop that fires every `--tick-secs` seconds (default `30`). Each pass does two things: it syncs configured boards/relays (pulling new offers and inbound sealed mes-

sages) and it runs the engine `tick`, which advances every live swap's state machine — funding when due, redeeming when the secret is available, fee-bumping a stuck transaction, and broadcasting a refund when a timelock deadline passes. This is the mechanism that makes swaps *unattended*: once a swap is underway, the scheduler will redeem your winnings and refund your losses without any further RPC calls.

Setting `--tick-secs 0` disables the loop; the engine then advances only when you call `tick` (or run with `--once`). That is useful for tests and for externally orchestrated setups, but for a normal deployment leave the scheduler running — a disabled scheduler means **no auto-refund**, which can lose funds if a counterparty disappears.

> **Warning** — Do not run a mainnet maker with `--tick-secs 0` and walk away. The auto-refund safety net only fires on scheduler ticks; with the loop off, a stalled swap will sit unrefunded until you manually tick it.

5.5 Example invocations

A regtest daemon that auto-creates its seed and runs over a local Corkboard:

```
pactd --data-dir ./data \
      --network regtest \
      --auto-init \
      --coin btcx=http://user:pass@127.0.0.1:19443/wallet/swap \
      --coin btc=http://user:pass@127.0.0.1:18443/wallet/swap \
      --board-url http://127.0.0.1:9780
```

A more realistic mainnet-shaped daemon: nested merchant layout, an encrypted seed opened from the environment, the default Nostr transport, and an explicit per-coin confirmation override:

```
PACT_PASSPHRASE='…' pactd \
      --data-dir /var/lib/pact \
      --network mainnet \
      --merchants \
      --coin btcx=http://user:pass@127.0.0.1:9332/wallet/swap \
      --coin btc=http://user:pass@127.0.0.1:8332/wallet/swap \
      --coin-confs btc=6 --coin-confs btcx=10 \
      --nostr-relay wss://relay.example.com,wss://relay2.example.com \
      --tick-secs 30
```

Here `--merchants` puts each seed under `merchants/<id>/`, `PACT_PASSPHRASE` unlocks the encrypted seed at boot, the two `--coin` flags attach wallet-qualified Core-RPC backends that fund the swap legs, and `--coin-confs` overrides the per-coin finality depth. The defaults for coins, pairs, and confirmation depth are explained in the next chapter.

Chapter 6

Coins, Pairs & Capabilities

Pact trades between *coins*. A coin is a Bitcoin-Core-compatible chain the engine knows how to verify, derive keys for, and build swap outputs on. Trading *pairs* are not curated — they are **derived** from what each pair of coins can do. This chapter covers the built-in coins, adding coins as data, attaching backends with `--coin`, setting confirmation depths, and how capabilities turn into pairs.

6.1 Built-in coins

Two coins ship in the engine code itself, fully trusted: `btcx` (Bitcoin-PoCX) and `btc` (Bitcoin). These are the engine's *registry* coins — their consensus parameters are compiled in and cannot be redirected by a stray file.

6.2 Adding coins with `coins.toml`

Any Bitcoin-Core-compatible chain can be added **as data**, with no recompile, through a `coins.toml` passed with `--coins-file`. The file is loaded at startup and merged with the built-ins; a file coin whose id collides with a built-in is dropped (so a stray file can never redirect `btc` or `btcx`), and a malformed file logs an error and falls back to the built-ins rather than refusing to boot.

Only the **consensus** fields the engine needs are read — enough to verify genesis, build and parse addresses, and derive keys. A minimal entry looks like:

```toml
schema_version = 1

[[coin]]
coin_id        = "ltc"
display_name   = "Litecoin"
symbol         = "LTC"
decimals       = 8
bip32_coin_type = 2
```

```
target_spacing_secs = 150

[coin.capabilities]
cltv      = true
segwit_v0 = true
taproot   = true

[coin.mainnet.consensus]
genesis_hash = "12a765e31ffd4059bada1e25190f6e98c99d9714d334efa41a195a7e7e04bfe2"
bech32_hrp   = "ltc"

[coin.regtest.consensus]
genesis_hash = "530827f38f93b43ed12af0b3ad25a288dc02ed74d6d7857862df51fc56c416f9"
bech32_hrp   = "rltc"
```

> **Note** — The consensus values in `coins.toml` are trusted by whoever edits the file, but they are not taken on faith at runtime: before any funds move, the engine validates each coin's `genesis_hash` against the live node (`getblockhash 0`). A wrong genesis hash fails the chain check, not a swap. A `coins.toml` entry may also carry a `connection` sub-table with RPC defaults; that is Satchel's concern and is ignored by the engine.

6.3 Attaching backends with `--coin`

A coin in the registry is *configured* by attaching a chain backend with `--coin id=url[,url]`:

- The **first URL is the wallet-qualified Core-RPC primary** — the bitcoind-style RPC, including the wallet path, that actually funds and broadcasts swap transactions. For example: `--coin btcx=http://user:pass@127.0.0.1:9332/wallet/swap`.
- Any **additional URLs may be Electrum** backends (`tcp://` or `ssl://`), used for light chain queries.

The coin id must already be in the registry (built-in or `coins.toml`-added); the flag is repeatable, once per coin, and the last `--coin` for a given id wins. This is the single, uniform way every coin is wired — there is no per-coin special-casing.

6.4 Confirmation depth with `--coin-confs`

Each coin has a *confirmation depth* `N` — how many confirmations a funding or redeem needs before the engine treats it as final. This gates auto-redeem and swap completion in both v1 and v2, and it is your reorg-safety knob. Override it per coin with `--coin-confs id=N` (`N ≥ 1`); coins you don't override use a default heuristic based on the network and the chain's block spacing:

Chain profile	Default `N`
Regtest (any coin)	`1`
Fast chain — block spacing under 5 minutes (e.g. BTCX, 120 s)	`10`
Slow chain — block spacing 5 minutes or more (e.g. BTC, 600 s)	`6`

Note — The fast-chain default is deliberately higher than the slow-chain one: a faster chain produces more blocks in the same wall-clock reorg window, so it needs more confirmations to reach comparable finality. Whatever you set, the floor is `1`.

6.5 Capabilities and how pairs are derived

Each coin declares three capabilities: `cltv` (OP_CHECKLOCKTIMEVERIFY, i.e. usable timelocks), `segwit_v0` (P2WSH outputs), and `taproot` (P2TR / v1 segwit outputs). These capabilities determine which *protocols* a pair of coins can use, and therefore which pairs are tradable at all — the engine **derives** the pair list, it never curates one:

- A pair where both coins have `cltv` + `segwit_v0` can run the **v1 HTLC** protocol.
- A pair where both coins have `taproot` can run the **v2 adaptor** protocol.

When both protocols are possible, the engine **prefers HTLC**; the v2 adaptor protocol is selected only for Taproot pairs that lack an HTLC option. The shipped BTCX ↔ BTC pair therefore defaults to HTLC. (Note that v2 adaptor swaps are enabled on every network, including mainnet.)

6.6 Inspecting coins and pairs

Three RPC methods report the live state of your coin and pair configuration — covered in full in the API part of this handbook, but worth knowing here:

- `listcoins` — every registry coin with its capabilities, whether it is `configured`, a live `status` probe, tip height, genesis hash, bech32 HRP, and its effective and default confirmation depth.
- `listpairs` — every derivable pair with its protocol list (`htlc` / `adaptor`), whether both legs are configured, and whether it is currently available.
- `validatecoin` — a genesis check of a *proposed* backend for a coin, without touching the engine's live configuration; use it to confirm a node before you attach it.

See the chapter on coins-and-pairs RPCs for the full field lists and return shapes.

Chapter 7

Seeds, Wallets & Merchants

Every key Pact ever uses — identity, swap keys, preimages, adaptor secrets, refund keys — is derived from a single BIP39 *seed*. This chapter covers the seed's lifecycle (create, import, encrypt, unlock), the encrypted-seed format, the *merchant* model (one seed per data directory), and what lives on disk.

7.1 The seed lifecycle

A data directory starts with **no seed**. From there, exactly one seed is established, by one of these routes:

- `createseed` — generate and persist a fresh seed; the mnemonic is returned **once** and never again.
- `importseed` — install an existing BIP39 mnemonic.
- `--auto-init` — at boot, `pactd` creates the seed and state on first run (flat layout) if none exists. Used by launchers like Satchel.
- `PACT_PASSPHRASE` — supplies the passphrase to *open an encrypted seed* at boot; it does not create one.

Once established, the seed is either **plaintext** (the regtest intent) or **encrypted** — created with a passphrase, in which case it lands *locked* and stays unusable until you `unlock` it (or supply `PACT_PASSPHRASE` at boot).

The daemon reports exactly where it stands through `WalletStatus`:

```
{ "seed_exists": true, "encrypted": true, "locked": false }
```

`locked` is true precisely when the seed is `encrypted` *and* the passphrase is not currently held in memory. `unlock` verifies a passphrase by trial-decryption and then holds it in memory for the process lifetime; `walletstatus` and `getinfo` both surface these three flags.

> **Note** — A plaintext seed is appropriate for regtest and disposable test setups. For testnet or mainnet, create the seed with a passphrase so it is encrypted at rest, and supply `PACT_PASSPHRASE` (or call `unlock`) to open it.

7.2 The encrypted seed format

The seed file is `seed.mnemonic`. In plaintext form it holds the BIP39 mnemonic directly. Encrypted, it is a single line:

`PACTSEEDv1:<salt>:<nonce>:<ciphertext>`

The mnemonic is encrypted with **ChaCha20-Poly1305** under a key derived by **scrypt** (`N = 2^15`, `r = 8`, `p = 1`) from the passphrase and per-file salt.

> **Warning** — Seed installation is non-overwriting and atomic: `install_seed` **refuses to overwrite** an existing seed, and writes via a temp file with fsync and rename so a crash mid-write cannot corrupt or truncate the seed. To replace a seed you must remove the data directory's seed deliberately — the engine will never clobber it for you.

7.3 The merchant model

A *merchant* is one identity backed by one seed in one data directory. Pact supports two layouts:

- **Flat (default).** A single seed lives in the data-dir root. This is the layout the harness and `pact-cli` rely on, and what `--auto-init` produces. It is internally a single `default` merchant.
- **Nested (`--merchants`).** Each merchant lives under `<data-dir>/merchants/<id>/`, with a `merchants.json` manifest in the parent. `pactd` owns an in-process registry; one merchant is *active* at a time, and the merchant RPCs (`createmerchant`, `listmerchants`, `loadmerchant`, `unloadmerchant`, `getmerchantinfo`) create and switch between them at runtime. This is the layout Satchel uses to manage several trading identities.

  > **Warning** — Switching identities has a fund-safety guard: `loadmerchant` and `unloadmerchant` **refuse to switch away from a merchant that has a live (non-terminal) swap**. This prevents you from orphaning a swap that still needs its scheduler ticks to redeem or refund. Finish, refund, or let the swap reach a terminal state before switching away.

The `--merchants` flag is ignored once a flat seed already exists in the data-dir root — an existing flat install stays flat.

7.4 What lives on disk

Per merchant data directory (the root in flat mode, `merchants/<id>/` in nested mode):

File	Contents
`pact.sqlite`	All swap, offer, nonce, and Nostr state (see below).

File	Contents
`seed.mnemonic`	The BIP39 seed — plaintext, or `PACTSEEDv1:…` if encrypted.
`.cookie`	The per-run RPC cookie (data-dir root only).
`pact.conf`	Optional `rpcuser` / `rpcpassword` for RPC auth.
`merchants.json`	The merchant manifest (parent data dir, nested mode only).

The SQLite database is the engine's durable state. At a high level its tables are: `swaps` and `adaptor_swaps` (the v1 and v2 swap records), `meta` (counters, the board `relay_cursor`, and private offers), `pending_takes`, `nonce_sessions` (the v2 MuSig2 use-once nonce state machine), `my_offers`, and the Nostr `nostr_outbox` / `nostr_inbox` / `nostr_offer_cache` tables.

> **Note** — Records are strict: a missing required field fails the load rather than silently defaulting. There is no serde-default migration path — this is a deliberate "no backward compatibility" stance, so do not hand-edit the SQLite state and expect old shapes to be tolerated.

Chapter 8

The pact-cli

`pact-cli` is the thin command-line client for `pactd` — the `bitcoin-cli` of this stack. It is a hand-rolled HTTP caller (a raw `TcpStream` with a 120-second read timeout and chunked-response decoding) that maps subcommands onto JSON-RPC methods. This chapter covers its global flags, the structured subcommand table, the generic `call` escape hatch (and which methods need it), and a worked end-to-end v1 swap.

8.1 Global flags

These apply to every invocation:

Flag	Default	Meaning
`--rpc`	`http://127.0.0.1:9737`	The `pactd` JSON-RPC endpoint.
`--data-dir`	—	Data directory to read the `.cookie` from (cookie auth).
`--rpcuser`	—	Explicit RPC username (overrides cookie auth).
`--rpcpassword`	—	Explicit RPC password.

Authentication uses the explicit `--rpcuser`/`--rpcpassword` credentials if given, otherwise the cookie read from `--data-dir`. This mirrors `bitcoin-cli`.

8.2 Subcommands

The CLI exposes structured subcommands for the common v1 and board operations:

Subcommand	Flags / args	RPC method
`call <method> [params...]`	each param JSON-parsed, else treated as a string	**any** (generic passthrough)
`offer`	`--give --get --t1 --t2 --out`	`offer`
`accept`	`--in --out`	`acceptoffer`
`recv`	`--in`	`recv`
`fund`	`--swap --out`	`fund`
`redeem`	`--swap`	`redeem`
`refund`	`--swap`	`refund`
`abort`	`--swap --reason`	`abort`
`status`	`--swap` (optional)	`getswap` / `listswaps`
`walletstatus`	—	`walletstatus`
`coins`	—	`listcoins`
`pairs`	—	`listpairs`
`validatecoin`	`--coin --backend`	`validatecoin`
`createseed`	`--passphrase` (optional)	`createseed`
`importseed`	`--mnemonic --passphrase` (optional)	`importseed`
`unlock`	`--passphrase`	`unlock`
`board post`	`--give --get --t1_secs` (default 12h) `--t2_secs` (default 6h)	`boardpostoffer`
`board offers`	—	`boardlistoffers`
`board take`	`--offer`	`boardtake`
`board revoke`	`--offer`	`boardrevoke`
`board sync`	—	`tick`

8.3 The generic `call` escape hatch

The `pact-cli call <method> [params...]` form invokes **any** RPC method directly. Each positional argument is JSON-parsed if it parses, otherwise passed as a string — so `call getbalance btc` and `call offer btcx:1.0 btc:0.9 ...` both work. This is how you reach everything that has no structured subcommand.

> **Note** — There is a real *CLI gap*: large parts of the RPC surface have no dedicated subcommand and are reachable **only** through `call`. That includes all of the **v2 adaptor** methods (`adaptorinit`, `adaptoraccept`, `adaptorfund`, `adaptorredeem`, …), all **merchant** methods (`createmerchant`, `loadmerchant`, …), the **private-offer** methods (`makeprivateoffer`, `takeoffer`, …), the **wallet** methods (`getbalance`, `getnewaddress`, `sendtoaddress`), and assorted others (`getinfo`, `estimateswapfees`, `generateseed`, `boardstatus`, `listmyoffers`, `listpendingtakes`). For those, use `call`.

For example, to drive a v2 adaptor swap or check a balance:

```
pact-cli call getinfo
pact-cli call getbalance btc
pact-cli call adaptorinit btcx:1.0 btc:0.95 86400 43200
pact-cli call estimateswapfees btcx btc
```

8.4 Worked example: a v1 swap with two CLIs

This is the manual happy path the harness drives in `test_swap_e2e.py`, with two daemons — **Alice** (the initiator, giving BTCX, getting BTC) and **Bob** (the participant). Each `pact-cli` here is assumed pointed at its own daemon's `--rpc` / `--data-dir`; messages are passed between the two as JSON files.

1. **Alice makes the offer.** She gives `btcx`, gets `btc`, with timelocks `t1` (her chain-A refund) and `t2 < t1` (Bob's chain-B refund). This writes an init envelope:

   ```
   alice> pact-cli offer --give btcx:1.0 --get btc:0.95 \
                         --t1 <T1> --t2 <T2> --out init.json
   ```

2. **Bob accepts.** He reads the init envelope, reconstructs the HTLC scripts locally, and emits an accept envelope:

   ```
   bob> pact-cli accept --in init.json --out accept.json
   ```

3. **Alice receives the acceptance** and funds her chain-A leg first (the maker funds first), producing a `funded` message that names the funding outpoint:

   ```
   alice> pact-cli recv --in accept.json
   alice> pact-cli fund --swap <swap_id> --out funded_a.json
   ```

4. **Bob verifies Alice's funding on-chain, then funds his chain-B leg:**

   ```
   bob> pact-cli recv --in funded_a.json
   bob> pact-cli fund --swap <swap_id> --out funded_b.json
   ```

5. **Alice receives Bob's funding, then redeems chain B** — this is the step that reveals her preimage `s` on the BTC chain:

   ```
   alice> pact-cli recv --in funded_b.json
   alice> pact-cli redeem --swap <swap_id>      # reveals s on chain B
   ```

6. **Bob redeems chain A.** His engine extracts `s` from Alice's chain-B spend and uses it to claim the BTCX leg:

   ```
   bob> pact-cli redeem --swap <swap_id>        # engine extracted s from chain B
   ```

Both HTLCs are now spent and the swap is `completed`. The `swap_id` is the 16-hex identifier returned by `offer` (and visible via `status`).

Tip — In a real deployment you do not perform steps 5 and 6 by hand. With the scheduler running (the default `--tick-secs 30`), each side's daemon redeems automatically once the required confirmations are reached, and auto-refunds if a deadline passes — *see the chapters "Running pactd" and the swap-protocol chapters*. The manual flow above is the way to understand and test the protocol, not the way to operate it.

Warning — The timelock ordering is load-bearing: `t2` (Bob's refund) must be earlier than `t1` (Alice's refund), and both must respect the network's action-deadline margins. The engine validates these offsets at offer time and will reject an unsafe pair; do not work around the rejection. See the timelock material in the protocol part of this handbook.

Part III

The JSON-RPC API

Chapter 9

JSON-RPC Conventions

`pactd` exposes the `libswap` engine over **JSON-RPC 2.0**. This chapter covers the transport, authentication, request/response envelope, and the common error shapes that every method shares. The per-method reference lives in the chapters that follow ("API: Node, Seed, Merchants, Coins", "API: v1 HTLC Swaps", "API: v2 Adaptor Swaps", and "API: Board, Private Offers & Fees").

> **Note** — `pactd` is to the swap engine what `bitcoind` is to Bitcoin: a long-running daemon driven entirely over RPC. `pact-cli` is the thin command-line client (the `bitcoin-cli` analog), and Satchel is the desktop GUI (the `bitcoin-qt` analog).

9.1 Transport

Property	Value
Protocol	JSON-RPC 2.0 over HTTP
Method endpoint	`POST /`
Health endpoint	`GET /health` (returns `ok`, **unauthenticated**)
Default listen address	`127.0.0.1:9737`
Binding	**Loopback only** — a non-loopback `--listen` aborts boot

All RPC calls are HTTP `POST` to the root path `/`. The daemon enforces that its listen address is a loopback address (`127.0.0.1`/`::1`); binding to a routable interface is rejected at startup. To reach `pactd` from another host, front it with your own authenticated tunnel — never expose the port directly.

`GET /health` is the only unauthenticated route. It returns the literal string `ok` and is intended for liveness probes.

9.2 Authentication

`pactd` uses **HTTP Basic** auth, exactly like `bitcoind`. Two credential sources are accepted (either works):

1. **Cookie (always written).** On startup `pactd` writes `<datadir>/.cookie` containing `__cookie__:<hex>`, where `<hex>` is a fresh 32-byte random value rendered as hex. The file is rewritten per run and removed on a clean shutdown. Use `__cookie__` as the username and the hex as the password.
2. **`pact.conf` (optional).** A `<datadir>/pact.conf` file with `key = value` lines (`#` introduces a comment). If it sets `rpcuser` and `rpcpassword`, those credentials are accepted **alongside** the cookie.

Both sources are normalized to a `Basic <base64(user:password)>` header and compared in constant time. `pact-cli` reads `.cookie` automatically when you pass `--data-dir`; pass `--rpcuser`/`--rpcpassword` to override.

> **Tip** — The passphrase that unlocks an encrypted seed is **not** an RPC credential. Supply it via the `unlock` method or the `PACT_PASSPHRASE` environment variable at boot. See the chapter "Wallet & Seed Lifecycle".

9.3 Request shape

Every request is a JSON object:

```
{ "jsonrpc": "2.0", "id": 1, "method": "getinfo", "params": [] }
```

Field	Required	Notes
`jsonrpc`	no	Ignored on input; the response always sets `"2.0"`.
`id`	no	Echoed verbatim in the response.
`method`	yes	One of the methods in the following chapters.
`params`	no	Defaults to `null`. See below.

Params accept a positional array OR a named object. These two calls are equivalent:

```
{ "method": "offer", "params": ["btcx:1.0", "btc:0.5", 144, 72] }
```

```
{ "method": "offer",
  "params": { "give": "btcx:1.0", "get": "btc:0.5", "t1": 144, "t2": 72 } }
```

Numeric params also accept numeric strings (e.g. `"144"`), matching `bitcoin-cli` behaviour. A missing required param fails with `missing param '<name>'` (positional form adds `(position <i>)`).

9.4 Response shape

A successful call returns:

```
{ "jsonrpc": "2.0", "id": 1, "result": { "...": "..." } }
```

An error returns an `error` object instead of `result`:

```
{ "jsonrpc": "2.0", "id": 1,
  "error": { "code": -1, "message": "unknown method 'frobnicate'" } }
```

Error	Message text
Unknown method	`unknown method '<method>'`
Missing param	`missing param '<name>'`
No merchant loaded	`no active merchant — create or load one first`

The error `code` is always `-1`; the human-readable detail is in `message`.

> **Note** — All swap, board, offer, and seed RPCs operate on the **active merchant's** engine. If no merchant is loaded (fresh nested-mode datadir with nothing selected), they fail with `no active merchant — create or load one first`. Use `createmerchant` / `loadmerchant` first. See the chapter "API: Node, Seed, Merchants, Coins".

9.5 Examples

A raw `curl` call against the cookie file:

```
# Cookie file holds "__cookie__:<hex>"; pass it straight to --user.
curl -s --user "$(cat ~/.pact/.cookie)" \
  -H 'content-type: application/json' \
  -d '{"jsonrpc":"2.0","id":1,"method":"getinfo","params":[]}' \
  http://127.0.0.1:9737/
```

The same call through `pact-cli` (which reads `.cookie` from `--data-dir`):

```
pact-cli --data-dir ~/.pact call getinfo
```

`pact-cli call <method> [params...]` is a generic passthrough: each argument is parsed as JSON if possible, otherwise treated as a string. It reaches **any** RPC method, including those without a dedicated subcommand.

> **Note** — `platform_fee_sat` is always `0`. There are no platform fees anywhere in the engine; the field exists only so fee previews have a consistent shape. See the chapter "API: Board, Private Offers & Fees".

Chapter 10

API: Node, Seed, Merchants, Coins

This chapter documents the non-swap RPC surface: node introspection, the seed lifecycle, the merchant model, the coin/pair registry, and the wallet helper methods. Conventions (transport, auth, request/response shape, the *no active merchant* error) are covered in the chapter "JSON-RPC Conventions".

10.1 Node / info

Method	Params	Returns	Mutates
`getinfo`	—	`{ name, version, protocol, network, identity?, seed_exists, encrypted, locked, coins }`	no
`walletstatus`	—	`{ seed_exists, encrypted, locked }`	no
`stop`	—	`"pactd stopping"`	yes (lifecycle)

- `getinfo` — `name` is always `"pactd"`; `version` is the crate version; `protocol` is the swap protocol version; `network` is the lowercased network name (`regtest`/`testnet`/`mainnet`); `coins` is the list of configured coin ids. Tolerates a missing or locked seed — `identity` is `null` until a seed is present **and** unlocked.
- `walletstatus` — the seed state triple. `locked` is true only when the seed is encrypted **and** its passphrase is not held in memory.
- `stop` — requests a graceful shutdown and returns immediately.

10.2 Seed lifecycle

Method	Params	Returns	Mutates
`createseed`	`passphrase?`	`{ mnemonic, encrypted }`	yes (writes seed)
`generateseed`	—	`{ mnemonic }`	no
`importseed`	`mnemonic`, `passphrase?`	`{ mnemonic, encrypted, identity }`	yes (writes seed)
`unlock`	`passphrase`	`{ unlocked, identity }`	yes (in-memory)

- `createseed` — generates and **persists** a new BIP39 seed. The `mnemonic` is returned exactly once, for the user to back up. `encrypted` is true iff a non-empty `passphrase` was supplied.
- `generateseed` — generates a mnemonic but does **not** persist it. Used by the onboarding flow to preview-then-confirm a phrase before committing it with `importseed`.
- `importseed` — installs a supplied `mnemonic` (optionally encrypted under `passphrase`). Echoes the normalized phrase plus the derived `identity` (npub-style pubkey). Refuses to overwrite an existing seed.
- `unlock` — verifies `passphrase` by trial-decrypt and holds it in memory for the process lifetime, returning the derived `identity`.

  Warning — A persisted mnemonic is shown only once by `createseed` / `importseed`. Back it up immediately; there is no recovery path if it is lost.

10.3 Merchants

A *merchant* is one seed bound to one data directory — the engine's wallet analog. The RPC surface is merchant-scoped: all swap/board/seed calls target the **active** merchant. Nested mode (`--merchants`) lays out `merchants/<id>/`; the flat layout has a single seed in the data-dir root.

Method	Params	Returns	Mutates	Mode
`createmerchant`	`label?`	`{ id, label }`	yes	nested only
`listmerchants`	—	`{ merchants:[…], active }`	no	any
`loadmerchant`	`id`	`{ id, label }`	yes	any
`unloadmerchant`	—	`{ unloaded }`	yes	nested only
`getmerchantinfo`	`id?`	merchant metadata	no	any

- `createmerchant` — allocates the next free id (`m<N>`) and makes it active.
- `listmerchants` — each entry is `{ id, label, identity?, created, encrypted, active, locked }`; `active` names the currently selected id.
- `loadmerchant` — switches the active merchant in-process.
- `unloadmerchant` — clears the active merchant.
- `getmerchantinfo` — metadata for one merchant, defaulting to the active one.

Warning — `loadmerchant` and `unloadmerchant` **refuse** to switch away from a merchant that has a live (non-terminal) swap, so an in-flight swap is never orphaned. Drive the swap to a terminal state first.

10.4 Coins / pairs

Method	Params	Returns	Mutates
`listcoins`	—	`{ network, coins:[…] }`	no
`listpairs`	—	`{ network, pairs:[…] }`	no
`validatecoin`	`coin_id`, `chain_data`	`{ ok, tip_height, genesis_hash? }`	no

`listcoins` returns every coin in the shipped registry that is defined on the active network. Each entry:

Field	Meaning
`id`	Canonical coin id (e.g. `btcx`, `btc`, `ltc`).
`display_name`	Human name.
`symbol`	Ticker.
`decimals`	Smallest-unit precision.
`capabilities`	`{ cltv, segwit_v0, taproot }` booleans.
`configured`	True if a chain backend is wired for this coin.
`status`	Live probe: `"ok"`, `"unconfigured"`, or `"error: …"`.
`tip_height`	Chain tip from the probe (`null` if unconfigured/errored).
`genesis_hash`	Expected genesis hash for this network.
`bech32_hrp`	Address HRP.
`confirmations`	Effective confirmation depth in force.
`default_confirmations`	The network/spacing default depth.

- `listpairs` — derived (never curated). Each `PairInfo` is `{ coin_a, coin_b, protocols, selectable?, both_configured, available }`, where `protocols` lists `htlc` and/or `adaptor`.
- `validatecoin` — genesis-hash checks a *proposed* backend (`chain_data`) before Satchel saves it. Builds an ephemeral backend; the running engine config is untouched.

10.5 Wallet helpers

Method	Params	Returns	Mutates
`getbalance`	`chain`	`{ balance_sat }`	no
`getnewaddress`	`chain`	`{ address }`	yes (advances HD index)
`sendtoaddress`	`chain`, `address`, `amount`	`{ txid }`	yes (broadcasts)

`chain` is a coin id (e.g. `btc`). `amount` for `sendtoaddress` is a decimal string in whole coin units. `getnewaddress` advances the HD derivation index; `sendtoaddress` constructs and broadcasts a payment.

Chapter 11

API: v1 HTLC Swaps

This chapter documents the **v1 (HTLC)** swap RPCs. These drive a classic hashlock/timelock atomic swap between two coins. The cross-chain mechanics — why two timelocks, what each transaction does — are covered in the protocol chapters; here we document the RPC surface only. For shared conventions see the chapter "JSON-RPC Conventions". For v2 (Taproot/MuSig2 adaptor) swaps see the chapter "API: v2 Adaptor Swaps".

> **Note** — *v1 (HTLC) and v2 (Taproot/MuSig2 adaptor) swaps are reviewed, audited, and live on mainnet.*

11.1 Swap state machine

A v1 swap is a `SwapRecord` whose `state` advances through this enum (snake_case in JSON):

`created → accepted → funded_a → funded_b → redeemed_b → completed`

State	Meaning
`created`	Offer created by the initiator; no on-chain action yet.
`accepted`	Counterparty accepted; both sides committed to terms.
`funded_a`	The first HTLC leg is funded on-chain.
`funded_b`	The second HTLC leg is funded on-chain.
`redeemed_b`	The B-side HTLC has been redeemed (secret revealed).
`completed`	Both legs settled; swap finished successfully.

Terminal failure states:

State	Meaning
`refunded`	A funded leg timed out and was refunded.
`aborted`	Swap cancelled before our HTLC funded.

11.2 The `coin:amount` convention

`give` and `get` are strings of the form `coin:amount`, where `coin` is a registry coin id and `amount` is a decimal in whole coin units — for example `"btcx:1.0"` and `"btc:0.5"`. `t1`/`t2` are the two relative timelocks in blocks (initiator and counterparty legs respectively).

11.3 Read methods

Method	Params	Returns	Mutates
`listswaps`	—	`[SwapRecord]`	no
`getswap`	`swap_id`	one `SwapRecord`	no
`listpendingtakes`	—	outstanding takes awaiting maker init	no
`listmyoffers`	—	`[{ offer_id, offer, state, created, valid_for, current_expiry, final_expiry, now }]`	no

- `listswaps` — every v1 swap record for the active merchant.
- `getswap` — a single record by `swap_id`.
- `listpendingtakes` — takes that have arrived but for which the maker has not yet initiated a swap (no `SwapRecord` exists yet — the UI's "initiating" pre-swaps).
- `listmyoffers` — the maker's own offers (the My-offers view). `current_expiry` is the rolling expiry (last refresh + relay TTL, capped at `final_expiry`); `final_expiry` is the maker-set hard expiry (`created + valid_for`); `now` is the server timestamp for client-side countdown rendering.

11.4 Lifecycle methods

Method	Params	Returns	Mutates	Purpose
`offer`	`give`, `get`, `t1`, `t2`	`{ record, envelope }`	yes	Start a swap as initiator.
`acceptoffer`	`envelope`	`{ record, envelope }`	yes	Accept a received offer envelope.
`recv`	`envelope`	`{ record }`	yes	Ingest a counterparty reply envelope.

Method	Params	Returns	Mutates	Purpose
`fund`	`swap_id`	`{ record, envelope }`	yes (broadcasts)	Broadcast our HTLC funding tx.
`redeem`	`swap_id`	`{ record }`	yes (broadcasts)	Redeem the counterparty HTLC (reveals secret).
`refund`	`swap_id`	`{ record }`	yes (broadcasts)	Reclaim our funded HTLC after timeout.
`abort`	`swap_id`, `reason?`	`{ record }`	yes	Cancel an unfunded swap.
`tick`	—	`{ events:[…] }`	yes	Advance the scheduler one pass.

- `offer` — initiates a swap with the given terms and returns the signed `envelope` to hand to the counterparty.
- `acceptoffer` — accepts a counterparty's offer `envelope`, returning a reply `envelope`.
- `recv` — ingests a counterparty envelope (e.g. an acceptance reply).
- `fund` — builds and broadcasts our HTLC funding transaction.
- `redeem` — spends the counterparty's funded HTLC, revealing the preimage.
- `refund` — reclaims our own funded HTLC once its timelock has expired.
- `abort` — cancels the swap; `reason` defaults to `"user aborted"`.
- `tick` — runs one scheduler pass (board sync + engine tick) and returns the resulting `events`, each `{ swap_id, action, detail }`.

  Warning — `abort` is **refused once our HTLC has funded**. After funding, the only safe exits are `redeem` (if you can claim the counterparty's leg) or `refund` (after your timelock expires). Aborting a funded swap is not an option because the coins are already committed on-chain.

Chapter 12

API: v2 Adaptor Swaps

This chapter documents the **v2 (Taproot/MuSig2 adaptor)** swap RPCs. A v2 swap settles on the cooperative key-path (indistinguishable from an ordinary Taproot spend) and uses a MuSig2 two-of-two with an adaptor signature instead of an on-chain hashlock. The cryptographic rationale — why an adaptor, how the MuSig2 nonce exchange binds the secret — is covered in the protocol chapter "v2 Taproot/MuSig2 Adaptor Swaps". For shared RPC conventions see the chapter "JSON-RPC Conventions"; for the simpler HTLC route see "API: v1 HTLC Swaps".

> **Note** — v2 adaptor swaps are **live on every network, including mainnet**. *v1 (HTLC) and v2 (Taproot/MuSig2 adaptor) swaps are reviewed, audited, and live on mainnet.*

12.1 Swap state machine

A v2 swap is an `AdaptorSwapRecord` whose `state` advances through this enum (snake_case in JSON):

```
created → accepted → nonces_exchanged → signed
        → funded_a → funded_b → redeemed_b → completed
```

State	Meaning
`created`	Initiator created the adaptor swap; terms set.
`accepted`	Counterparty accepted the terms.
`nonces_exchanged`	MuSig2 nonces exchanged for both legs.
`signed`	Adaptor + partial signatures produced.
`funded_a`	The first leg's funding tx is confirmed.
`funded_b`	The second leg's funding tx is confirmed.
`redeemed_b`	The B-side has been redeemed (adaptor secret revealed).
`completed`	Both legs settled successfully.

Terminal failure states: `refunded`, `aborted`.

12.2 Handshake order

The methods below are **not** interchangeable — they must be driven in this order to complete the MuSig2 / adaptor handshake:

```
adaptorinit → adaptoraccept → adaptorrecv → adaptorfundingready
            → adaptornonces → adaptorsign → adaptorassemble
            → adaptorfund → adaptorredeem | adaptorrefund
```

`init`/`accept`/`recv` establish terms; `fundingready` registers the funding outpoint; `nonces`/`sign`/`assemble` complete the MuSig2 adaptor exchange; `fund` broadcasts; and the swap settles with `redeem` (cooperative path) or `refund` (timeout path). See "v2 Taproot/MuSig2 Adaptor Swaps" for why each step is required.

> **Note** — The cooperative key-path **redeem is NOT RBF-bumpable**; it is handled by fee over-provisioning plus a CPFP child. The single-key **refund IS bumpable**. Budget fees accordingly when settling near a deadline.

12.3 Read method

Method	Params	Returns	Mutates
`listadaptorswaps`	—	`[AdaptorSwapRecord]`	no

12.4 Handshake & lifecycle methods

`give`/`get` use the same `coin:amount` string convention as v1 (e.g. `"btcx:1.0"`); `t1`/`t2` are the two relative timelocks in blocks.

Method	Params	Returns	Mutates
`adaptorinit`	`give`, `get`, `t1`, `t2`	`{ record, envelope }`	yes
`adaptoraccept`	`envelope`	`{ record, envelope }`	yes
`adaptorrecv`	`envelope`	`{ record }`	yes
`adaptorfundingready`	`swap_id`, `txid`, `vout`	`{ envelope }`	yes
`adaptornonces`	`swap_id`	`{ envelope }`	yes
`adaptorsign`	`swap_id`	`{ envelope }`	yes
`adaptorassemble`	`swap_id`	`{ record }`	yes
`adaptorfund`	`swap_id`	`{ envelope }`	yes (broadcasts)
`adaptorredeem`	`swap_id`	`{ record }`	yes (broadcasts)
`adaptorrefund`	`swap_id`	`{ record }`	yes (broadcasts)

- `adaptorinit` — initiate a v2 swap with the given terms; returns the offer `envelope`.

- `adaptoraccept` — counterparty accepts; returns a reply `envelope`.
- `adaptorrecv` — ingest a counterparty envelope into the local record.
- `adaptorfundingready` — register the funding outpoint (`txid`, `vout`) once the funding UTXO exists; returns an envelope for the counterparty.
- `adaptornonces` — produce and exchange the MuSig2 nonces.
- `adaptorsign` — produce the adaptor and partial signatures.
- `adaptorassemble` — assemble the partials into the final signature set.
- `adaptorfund` — broadcast our funding transaction.
- `adaptorredeem` — settle on the cooperative key-path, revealing the adaptor secret.
- `adaptorrefund` — reclaim our funded leg via the single-key timeout path.

Chapter 13

API: Board, Private Offers & Fees

This chapter documents the board (Corkboard / Nostr) RPCs, the private ("trade-with-a-friend") offer RPCs, and the fee-estimation method. For shared conventions see the chapter "JSON-RPC Conventions"; for the swap lifecycle those offers feed into, see "API: v1 HTLC Swaps" and "API: v2 Adaptor Swaps".

13.1 Board (Corkboard / Nostr)

An offer posted to the board fans out to every configured transport (HTTP Corkboard URLs and/or Nostr relays). Reads are per-board: a `board` selector chooses one transport rather than merging.

Method	Params	Returns	Mutates
`boardlistoffers`	`board?`	`{ offers }`	no
`boardstatus`	—	`{ relays:[{ url, connected }] }`	no
`boardpostoffer`	`give`, `get`, `t1_secs`, `t2_secs`, `protocol?`, `ttl_secs?`	`{ offer_id }`	yes
`boardtake`	`offer_id`	`{ taken }`	yes
`boardrevoke`	`offer_id`	`{ revoked }`	yes

- `boardlistoffers` — lists offers from one board. The optional `board` is an HTTP Corkboard URL **or** the literal `"nostr"`; omitted, it defaults to the first configured board.
- `boardstatus` — relay connectivity for the header indicator: one `{ url, connected }` entry per configured Nostr relay. Empty when the Nostr transport is not configured.
- `boardpostoffer` — posts a listing. `give`/`get` are `coin:amount` strings; `t1_secs`/`t2_secs` are the two timelocks **in seconds**. `protocol` (param 4) optionally forces `"pact-htlc-v1"` or `"pact-htlc-v2"`; omitted, the engine picks the default for the pair. `ttl_secs` (param 5) sets the listing validity; omitted, the engine default applies.

- `boardtake` — takes a posted offer by `offer_id`.
- `boardrevoke` — revokes one of your own posted offers.

  Note — On Nostr the listing's expiry is **rolling**: `min(now + 1800s, created + ttl_secs)`. It is refreshed as the offer is republished, not pinned to `ttl_secs` from creation.

13.2 Private (off-market) offers

A private offer is built and signed locally but **never posted to a board**. It travels to a counterparty as a *slip* string (paste it into your own chat). The methods mirror the board ones, but no board is touched.

Method	Params	Returns	Mutates
`makeprivateoffer`	`give`, `get`, `t1_secs`, `t2_secs`, `protocol?`, `ttl_secs?`	`{ slip }`	yes
`takeoffer`	`slip`	`{ taken }`	yes
`listprivateoffers`	—	`{ offers }`	no
`cancelprivateoffer`	`offer_id`	`{ cancelled }`	yes

- `makeprivateoffer` — builds and signs an offer, returning the `slip` string to hand to a friend. As with `boardpostoffer`, `protocol` is **param 4** and `ttl_secs` is **param 5** (both optional).
- `takeoffer` — takes a private offer from a pasted `slip`: decodes and verifies the signed offer, then relays a take to the maker.
- `listprivateoffers` — your outstanding private offers.
- `cancelprivateoffer` — cancels one by `offer_id`.

13.3 Fees

Method	Params	Returns	Mutates
`estimateswapfees`	`give_coin`, `get_coin`	fee preview (see below)	no

`estimateswapfees` previews the on-chain fees for a prospective swap. It takes **only** `give_coin` and `get_coin` (coin ids).

> **Warning** — Older docs list `protocol?` and `role?` params for `estimateswapfees`. Those are **not parsed** — passing them has no effect. Legs are keyed solely off `give`/`get` (you fund what you give, redeem what you get).

The result shape:

```
{
  "platform_fee_sat": 0,
  "give": {
    "coin_id": "btcx",
    "fee_rate_sat_per_vb": 5,
    "fee_rate_is_fallback": false,
    "legs": [
      { "name": "fund",   "vbytes": 0, "fee_sat": 0 },
      { "name": "refund", "vbytes": 0, "fee_sat": 0 }
    ]
  },
  "get": {
    "coin_id": "btc",
    "fee_rate_sat_per_vb": 5,
    "fee_rate_is_fallback": false,
    "legs": [
      { "name": "redeem", "vbytes": 0, "fee_sat": 0 }
    ]
  }
}
```


- `platform_fee_sat` is always `0` — the board takes nothing.
- The `give` side covers the `fund` and `refund` legs; the `get` side covers the `redeem` leg. Each leg reports `{ name, vbytes, fee_sat }`.
- `fee_rate_is_fallback` is true when a live fee estimate was unavailable and a built-in fallback rate was used.

Part IV

The Swap Protocol

Chapter 14

The Swap Lifecycle

Pact settles a trade as an *atomic swap*: two on-chain transactions, one per chain, bound so that either both parties get paid or both get their money back. Nobody can run off with the funds. The chains — not a custodian, not a relay, not Pact itself — enforce the deal. This chapter introduces the two state machines that drive a swap to completion (v1 HTLC and v2 adaptor), the roles each party plays, and the scheduler that walks every live swap forward on a clock.

Two protocol versions ship today, both running on every network under external audit:

- **v1 (`pact-htlc-v1`)** — classic hash-timelock contracts (HTLCs) over P2WSH outputs. See the chapter "v1 HTLC Swaps".
- **v2 (`pact-htlc-v2`)** — Taproot outputs with a MuSig2 adaptor-signature binding. See the chapter "v2 Taproot/MuSig2 Adaptor Swaps".

The lifecycle, roles, and scheduler model described here are shared by both; the difference is what the on-chain commitment *looks like* and how the secret that unlocks it is revealed.

14.1 The two roles

Every swap has exactly two parties, and the roles are fixed for the duration:

Role	Code name	Who	Locks	Refund deadline	Holds the secret
Initiator	`Initiator` (Alice)	the maker — posts the offer	chain **A**	`T1` (later)	yes
Participant	`Participant` (Bob)	the taker — accepts the offer	chain **B**	`T2 < T1` (earlier)	no

`Role` is defined in `swap.rs:50-57` (v1) and the same Alice-funds-A / Bob-funds-B convention is documented in `adaptor_swap.rs:5-8` (v2).

The asymmetry is the whole point of the design:

- **The maker funds first.** Alice locks her coin on chain A before Bob locks his on chain B. The party who reveals the secret (Alice) is the one who must commit first, so she cannot strand Bob's funds without exposing her own.
- **Alice holds the secret.** In v1 it is a SHA256 preimage `s`; in v2 it is an adaptor scalar `t`. Either way, only the initiator can derive it from her seed, and revealing it on chain B is what lets Bob claim chain A.
- **`T2 < T1`.** Bob's refund unlocks *before* Alice's. Bob must always have a safe window to refund leg B after Alice's window to redeem it closes, before his own funds are at risk on the other side. This single inequality is the structural backbone of every Pact timelock; see the chapter "Timelocks & Action Deadlines".

  Note — "Maker funds first" is a protocol invariant, not a UI choice. The chain enforces the deal once both legs are funded; until then, the only party with capital at risk is the one who controls when the secret is revealed.

14.2 The scheduler-driven model

A Pact swap is not a request/response RPC dance. Once a swap is `Accepted`, the daemon owns it: `pactd` runs a background scheduler that re-examines every live swap on a fixed interval and takes whatever action the current state and the chain clock allow. The operator does not poll, click "redeem", or watch the mempool — the engine does.

The loop is `Engine::tick`, which fans out to one handler per swap (`engine.rs:2581`). For v1 each swap is advanced by `tick_one` (`engine.rs:2654`); for v2 by `adaptor_tick_one` (`engine.rs:2614-2620`). The interval is set by the `--tick-secs` flag.

On each tick, for each swap, the engine:

1. Reads the current state and the party's role.
2. Checks the chains it watches: have the funding outputs confirmed? has the counterparty redeemed and thereby revealed the secret? has a refund timelock matured (measured against the chain's median-time-past)?
3. Arms exactly the action the state machine permits — fund, redeem, refund, fee-bump, or nothing — and persists the new state.

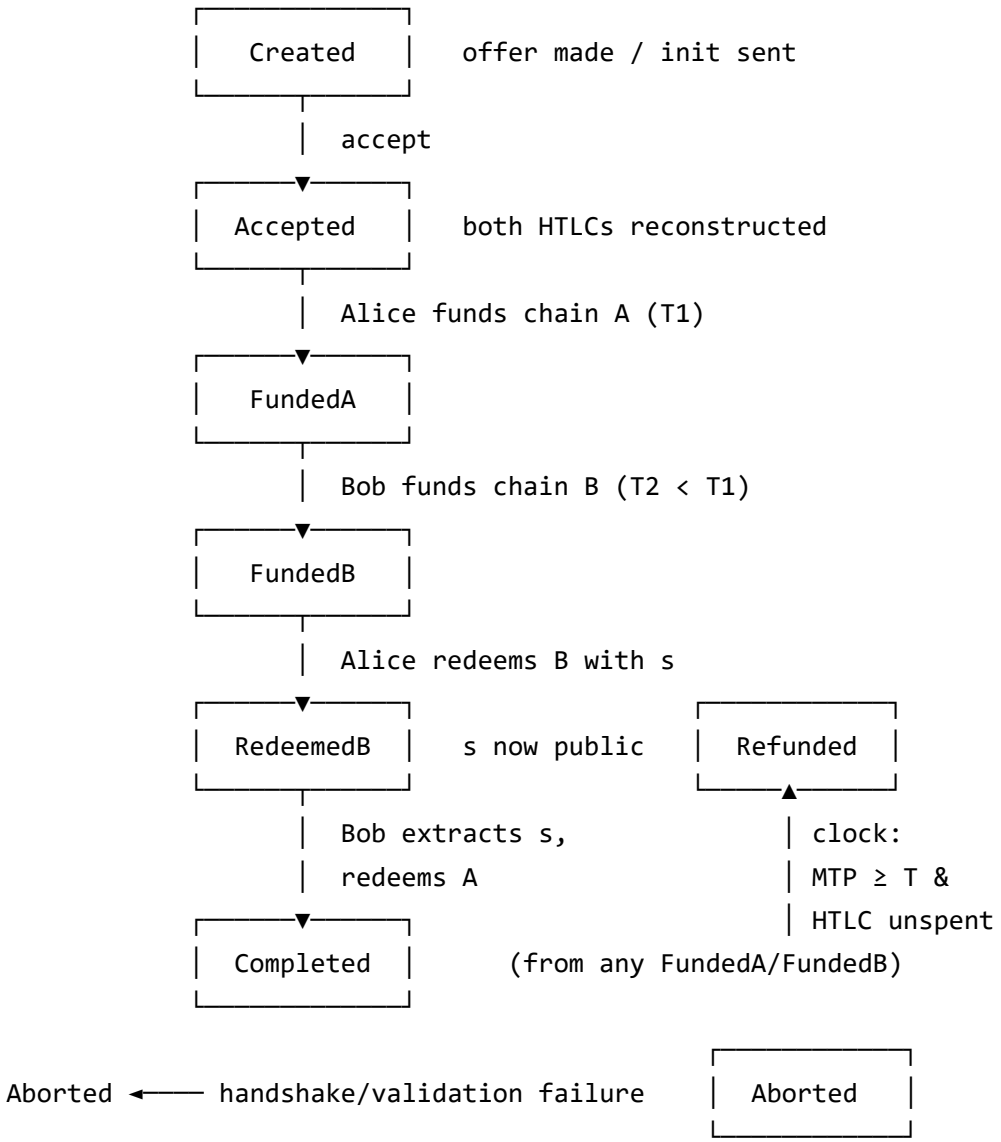
This is what makes **auto-redeem and auto-refund clock-driven**. Alice's daemon redeems leg B the moment it is safe; Bob's daemon extracts the revealed secret and redeems leg A; and either party's daemon fires the timelock refund the moment its deadline passes and the output is still unspent — all without human intervention, and all while the operator is offline-tolerant within the timelock margins (see the chapter "Timelocks & Action Deadlines"). The per-role, per-state arming lives in `engine.rs:2666-2774` for v1.

> **Warning** — Because the engine acts on a clock, a daemon that is *down* across a critical deadline cannot fire its refund. The action-deadline margins (fund 3h / reveal 2h / redeem-A 1h on mainnet) exist precisely to give a recovering daemon slack. Keep

`pactd` running for the life of any swap you have funded. See the chapter "Timelocks & Action Deadlines".

14.3 v1 state machine — `swap::State`

The v1 lifecycle is `swap::State` (`swap.rs:61-72`). The happy path is a straight line; the two refund/abort states branch off from any funded state, driven by the clock rather than by a message.

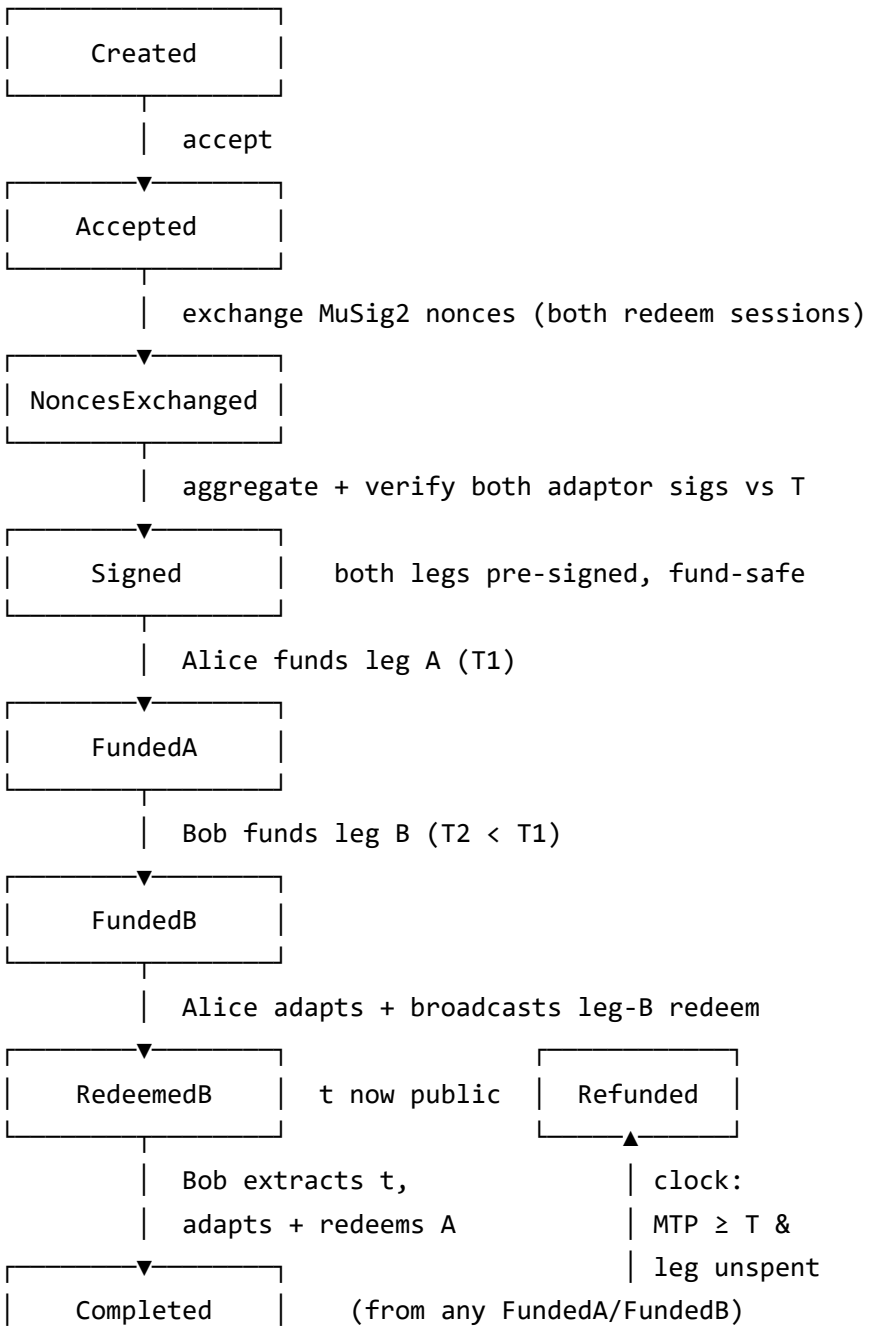


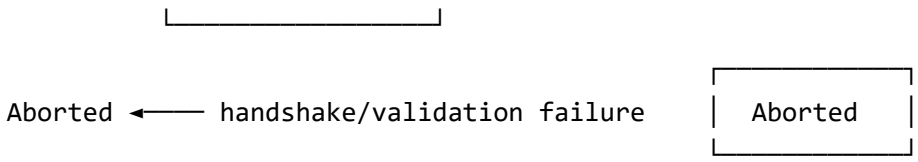
- `Created → Accepted`: the offer is taken; both parties now hold enough parameters (`SwapParams`, `swap.rs:74-91`) to reconstruct **both** HTLCs byte-for-byte and verify the counterparty's script locally.
- `Accepted → FundedA`: Alice broadcasts the chain-A funding transaction.
- `FundedA → FundedB`: Bob, having verified leg A on chain, funds leg B.
- `FundedB → RedeemedB`: Alice redeems leg B by revealing the preimage `s` in the spending witness. This is the irreversible step.

- `RedeemedB → Completed`: Bob's daemon scans the chain-B spend, extracts `s`, and redeems leg A.
- `Refunded` is reachable from `FundedA` or `FundedB` when a refund timelock matures and the corresponding HTLC is still unspent. `Aborted` covers pre-funding handshake or validation failures.

14.4 v2 state machine — `AdaptorState`

The v2 lifecycle is `AdaptorState` (`adaptor_swap.rs:34-48`). It is the same shape as v1 with **two extra pre-funding states** for the MuSig2 adaptor ceremony that must complete *before* either party puts coins on chain.





The two added states are the heart of v2's safety:

- `Accepted → NoncesExchanged`: both parties exchange fresh MuSig2 secret nonces for the two redeem sessions (one per leg). Nonces are generated from a CSPRNG, never the seed, and persisted write-ahead — reuse is structurally impossible. See the chapter "v2 Taproot/MuSig2 Adaptor Swaps".
- `NoncesExchanged → Signed`: both adaptor signatures are aggregated and **verified against the adaptor point `T`** before anyone funds. Reaching `Signed` is the guarantee that the cooperative redeems will work and that Alice's broadcast will reveal `t` to Bob.

  Note — In v2 the legs are pre-signed (state `Signed`) before any funding transaction is broadcast. By the time a party commits coins, the redeem messages already exist and have been checked against `T`. From `FundedB` onward, v2 behaves like v1: Alice's redeem reveals the secret on chain, Bob's daemon extracts it and claims his leg.

14.5 Where the secret lives

Both versions hinge on a single secret that only the initiator can produce, and both derive it from the seed so that losing the state database never loses the secret:

	v1	v2
Secret	preimage `s` (32 bytes)	adaptor scalar `t`
Public commitment	`H = SHA256(s)`, on both legs	point `T = t·G`, binds both redeem sigs
Revealed by	the redeem witness on leg B	the adapted leg-B redeem signature
Extracted by Bob	SHA256-scan of the leg-B witness	subtracting his adaptor sig from the broadcast sig

The mechanics of derivation, the exact witness layouts, and the extraction procedure are covered in the v1 and v2 chapters that follow. The lifecycle to hold in mind is the same in both: maker funds first, the chain enforces the deal, and the scheduler redeems or refunds on a clock.

Chapter 15

v1 HTLC Swaps

The v1 protocol (`pact-htlc-v1`) settles a trade with two *hash-timelock contracts* (HTLCs), one per chain. A single SHA256 preimage `s` unlocks both: the initiator reveals `s` to claim her leg, which publishes `s` on chain so the participant can claim his. Each leg also has a timeout branch that refunds the funder after an absolute Unix-time deadline. This chapter walks the protocol end to end — key derivation, the witness script, funding, redeem, refund, and preimage extraction — reproducing the exact byte-level constructions from `libswap`.

Roles are fixed (see the chapter "The Swap Lifecycle"): *Alice* is the `Initiator`, holds the preimage, locks chain A, and refunds at `T1`; *Bob* is the `Participant`, locks chain B, and refunds at `T2 < T1`. The maker funds first.

15.1 Key derivation

All key material derives from one BIP39 seed via BIP32, under the Pact *purpose* `7228'` (`7228` is "PACT" on a phone keypad, `keys.rs:20-21`). Every level is hardened.

Material	Path	Type & use
Identity key	`m/7228'/0'/0'`	BIP340 x-only; signs handshake envelopes. **Never** appears in an HTLC.
Swap key	`m/7228'/1'/coin(c)'/i'`	compressed secp256k1, ECDSA; the redeem/refund key. One per chain per swap.
Preimage source	`m/7228'/2'/i'`	feeds the preimage tagged hash (initiator only).

The `coin(c)` index identifies the **asset**, not the network (`keys.rs:23-25`):

Asset	coin(c)
BTC	0
PoCX (BTCX)	0x504F4358 (ASCII "POCX")

The preimage and the swap identifier are deterministic, so both survive loss of the state database (keys.rs:83-89, keys.rs:137-142):

```
s        = TaggedHash("pact/htlc/preimage/v1", privkey at m/7228'/2'/i')   (32 bytes)
H        = SHA256(s)                                                        (the hashlock)
swap_id  = hex( TaggedHash("pact/swapid/v1", H)[0..8] )                     (16 hex chars)
```

Only Alice can derive s (she owns the seed and the index i); Bob only ever learns H until Alice reveals s on chain.

> **Note** — The swap key is ECDSA in v1. The *same* derivation path is reused in v2 as a BIP340 x-only MuSig2 signer — same private key, different public encoding. See the chapter "v2 Taproot/MuSig2 Adaptor Swaps".

15.2 The HTLC witness script

Each leg is a P2WSH output committing to this witness script (htlc.rs:62-87). The script is identical on both chains; only the keys and the locktime T differ per leg.

```
OP_IF
    OP_SIZE 32 OP_EQUALVERIFY
    OP_SHA256 <H> OP_EQUALVERIFY
    OP_DUP OP_HASH160 <hash160(redeem_pubkey)>
OP_ELSE
    <T> OP_CHECKLOCKTIMEVERIFY OP_DROP
    OP_DUP OP_HASH160 <hash160(refund_pubkey)>
OP_ENDIF
OP_EQUALVERIFY
OP_CHECKSIG
```

- The **OP_IF (hash) branch** is the redeem path: present a 32-byte item whose SHA256 equals H, plus a signature from redeem_pubkey. The OP_SIZE 32 OP_EQUALVERIFY guard forbids the empty-push trick that would otherwise let OP_SHA256 run on a zero-length item.
- The **OP_ELSE (timeout) branch** is the refund path: after the absolute locktime T matures (enforced by OP_CHECKLOCKTIMEVERIFY against BIP113 median-time-past), a signature from refund_pubkey spends it.

The scriptPubKey is the standard P2WSH wrapper (htlc.rs:89-92):

```
OP_0 <SHA256(witness_script)>
```

15.2.1 Per-leg keys and locktimes

The two legs assign the redeem/refund roles in mirror image (`swap.rs:106-124`):

Leg	`redeem_pubkey`	`refund_pubkey`	`T`
Chain **A** (Alice funds)	`bob_redeem_a`	`alice_refund_a`	`t1`
Chain **B** (Bob funds)	`alice_redeem_b`	`bob_refund_b`	`t2`

On leg A, Bob redeems (with `s`) and Alice refunds at `T1`. On leg B, Alice redeems (with `s`) and Bob refunds at `T2`. Both legs share the same hashlock `H`.

> **Warning** — The locktime `T` MUST be a **Unix timestamp ≥ 500000000**; block-height locktimes are rejected at construction (`htlc.rs:28-29`, `htlc.rs:48-52`). HTLCs are time-locked, not height-locked, so a deadline means the same wall-clock instant on both chains regardless of their block cadence.

> **Warning** — A party MUST reconstruct the counterparty's witness script locally from the agreed `SwapParams` and **byte-compare** it before funding. Never trust an address or script handed to you over the wire — derive it.

15.3 Funding

Funding is an ordinary core-wallet send to the leg's exact-amount P2WSH address (spec §6.1). The funder's node performs coin selection, change, and fee estimation as for any payment; Pact only supplies the destination and value. The funder then sends a `funded(txid, vout)` message, and the receiver **verifies the output on chain** (correct script, correct amount, sufficient confirmations) rather than trusting the message. The engine can also discover funding by watching the chain directly if the message is withheld.

The funding vsize used for fee previews is `FUND_TX_VSIZE = 160` (`swap.rs:34`) — a 1-input / 2-output segwit send is a sensible mid-point; a real wallet selecting more inputs may differ.

15.4 Redeem

The redeem transaction spends the HTLC via the hash branch (`swap.rs:178-205`). It is a version-2, 1-in / 1-out transaction with `nLockTime = 0` and an RBF-signalling input sequence. The witness stack is:

```
[ sig‖0x01, redeem_pubkey (33), s (32), 0x01, witness_script ]
```

Item	Bytes	Purpose
`sig‖0x01`	DER + `SIGHASH_ALL` byte	BIP143 signature over the spend

Item	Bytes	Purpose
`redeem_pubkey`	33	the compressed redeem key, hashed in the script
`s`	32	the preimage; SHA256 must equal `H`
`0x01`	1	the `OP_IF` selector — a non-empty item takes the hash branch
`witness_script`	var	the full script, revealed for P2WSH

Key parameters (`swap.rs:127-205`):

- `nSequence = 0xFFFFFFFD` (`HTLC_SPEND_SEQUENCE`, `swap.rs:17`) — signals RBF so the redeem fee can be bumped (see the chapter "Fees, Fee-Bumping & Auto-Refund").
- `nLockTime = 0` — the hash branch has no timelock.
- Signature is BIP143 `SIGHASH_ALL` (`swap.rs:158-169`).
- Worst-case vsize for fee math: `REDEEM_TX_VSIZE = 155` (`swap.rs:25`).

The spend sweeps `htlc_value − fee` to the claimer's destination; the engine refuses to build a spend whose output would fall below the dust limit (`DUST_LIMIT_SAT = 546`, `swap.rs:19-20`, guarded at `swap.rs:138-141`).

15.5 Refund

The refund transaction spends the HTLC via the timeout branch (`swap.rs:210-235`). The witness stack uses an **empty** item to select the `OP_ELSE` branch:

`[ sig‖0x01, refund_pubkey (33), <empty>, witness_script ]`

Item	Bytes	Purpose
`sig‖0x01`	DER + `SIGHASH_ALL` byte	signature from the refund key
`refund_pubkey`	33	the compressed refund key
`<empty>`	0	the `OP_ELSE` selector — an empty item is false, taking the timeout branch
`witness_script`	var	the full script

Parameters:

- `nLockTime = T` — set to the leg's absolute locktime, so the transaction is only valid once the chain's median-time-past reaches `T`. Broadcasting earlier is rejected by the node (not fatal; the engine retries).

- nSequence = 0xFFFFFFFD — RBF-signalling, so the refund is also bumpable.
- Worst-case vsize: REFUND_TX_VSIZE = 146 (swap.rs:26).

  Note — The v1 refund is **signed and persisted at funding time** (rec.refund_tx_hex). The instant a party funds a leg, the fully-signed refund transaction already exists on disk, ready to broadcast the moment its timelock matures — even if the daemon has been restarted since. v2 differs: it re-derives the refund from the seed on each call. See the chapter "Fees, Fee-Bumping & Auto-Refund".

15.6 Preimage extraction

When Alice redeems leg B, her witness publishes s on chain B. Bob's daemon recovers it by scanning the spend's witness for a 32-byte item whose SHA256 equals the known H (htlc.rs:104-114):

```
for item in witness_items {
    if item.len() == 32 && SHA256(item) == hash_h {
        return Some(item);   // this is s
    }
}
```

The candidate is **verified against H**, never taken positionally on faith — backend data is untrusted, and a malicious node cannot feed Bob a bogus preimage that passes the SHA256 check. Once Bob has s, he redeems leg A with it before T1.

15.7 Message flow

The end-to-end handshake (spec §8) is:

1. init — Alice proposes the swap (amounts, H, T1/T2, her pubkeys).
2. accept — Bob agrees and supplies his pubkeys. Both sides can now reconstruct both HTLCs (SwapParams, swap.rs:74-91).
3. funded(A) — Alice funds chain A and announces it; Bob verifies on chain.
4. funded(B) — Bob funds chain B and announces it; Alice verifies on chain.
5. redeemed — Alice redeems leg B (revealing s); the courtesy message tells Bob, but his daemon would find it by chain-watching anyway.
6. Bob extracts s and redeems leg A. The swap is Completed.

From step 3 onward the daemon's scheduler drives the swap automatically: it funds, redeems, and (if a deadline passes with a leg unspent) refunds on a clock. See the chapters "The Swap Lifecycle" and "Timelocks & Action Deadlines".

15.8 vsize reference

Constant	Value (vB)	Transaction
`FUND_TX_VSIZE`	160	funding (preview estimate)
`REDEEM_TX_VSIZE`	155	hash-branch redeem
`REFUND_TX_VSIZE`	146	timeout-branch refund

These worst-case vsizes turn a feerate into an absolute fee before the witness exists. The fee floor for any HTLC spend is `MIN_SPEND_FEE_SAT = 1000` (`swap.rs:38`).

Chapter 16

v2 Taproot/MuSig2 Adaptor Swaps

The v2 protocol (`pact-htlc-v2`) settles the same trade as v1 but commits to each leg as a *Taproot* (P2TR) output instead of a hash-locked P2WSH script. The cooperative redeem is a single 64-byte Schnorr signature produced by a 2-of-2 MuSig2 key-path spend — on chain it is indistinguishable from an ordinary single-key payment. There is no preimage and no shared hash linking the two legs; instead a MuSig2 *adaptor signature* under a common point `T` binds them, and broadcasting one redeem reveals the scalar `t` that unlocks the other. The refund path is a single-key CLTV tapleaf — no MuSig2, no interactive nonce — so it is safe to run unattended.

Roles are unchanged from v1 (`adaptor_swap.rs:5-8`): Alice funds leg A (refund `T1`), Bob funds leg B (refund `T2 < T1`), Alice holds the adaptor secret `t`.

Both chains MUST support Taproot. PoCX has Taproot active from genesis; Bitcoin since block 709632.

16.1 Keys and secrets

v2 reuses the v1 paths but adds a refund-key branch and a new adaptor secret (`keys.rs:91-135`, spec v2 §3):

Material	Path	v2 type & use
Swap key	`m/7228'/1'/coin(c)'/i'`	secp256k1, used as a **BIP340 x-only key and MuSig2 signer** (was ECDSA in v1)
Refund key	`m/7228'/3'/coin(c)'/i'`	secp256k1 x-only; sole signer of the single-key CLTV refund tapleaf — a **new, separate** branch
Adaptor secret source	`m/7228'/2'/i'`	feeds the adaptor tagged hash (was the v1 preimage source)

The swap key is the *same private key* as v1, encoded x-only instead of compressed (`keys.rs:96-101`). The refund key being a distinct branch (`3'`) is deliberate: it keeps the refund tapleaf single-signature and independent of the MuSig2 aggregate, so the unattended refund path never needs an interactive ceremony.

The adaptor secret and the swap id are deterministic (`keys.rs:118-142`, spec v2 §3.1):

```
k  = privkey at m/7228'/2'/i'                                  (32 bytes)
t  = TaggedHash("pact/adaptor/secret/v2", k)  mod n            (a valid secp256k1 scalar, ≠ 0)
T  = t·G                                                       (the adaptor point, shared in in
swap_id = hex( TaggedHash("pact/swapid/v2", T)[0..8] )         (16 hex chars)
```

`t` is the v2 analog of v1's preimage `s`: Alice-only, seed-re-derivable, never disclosed off-chain. The point `T` is public and is what binds the two redeem signatures together.

16.2 The Taproot output

Each leg is one P2TR output (`taproot.rs:43-120`, spec v2 §4). Its two spend paths are:

- **Key path** = the 2-of-2 MuSig2 aggregate of both parties' swap keys — the cooperative redeem.
- **Script path** = a single tapleaf holding a single-key CLTV refund script.

The internal key `P` is the MuSig2 aggregate, with a **fixed key order `[funder, counterparty]`** (`adaptor_swap.rs:86-90`). The single refund tapleaf is (`taproot.rs:72-81`):

```
<T> OP_CLTV OP_DROP <funder_refund_xonly> OP_CHECKSIG
```

encoded as a TapScript leaf (leaf version `0xc0`). The output key is `P` tweaked by the leaf's merkle root (`taproot.rs:83-98`), and the address is bech32m.

The leg assignments mirror v1 (`adaptor_swap.rs:85-90`, and leg B symmetric):

Leg	MuSig2 internal key order	refund tapleaf key	`T`
A (Alice funds)	`[alice_swap_a, bob_swap_a]`	`alice_refund_a`	`t1`
B (Bob funds)	`[bob_swap_b, alice_swap_b]`	`bob_refund_b`	`t2`

> **Note** — Because the key path is a 2-of-2 MuSig2 spend, a successful v2 redeem is on-chain indistinguishable from any other single-key Taproot payment. There is no hash, no script, and nothing linking leg A to leg B visible to a chain observer. Privacy is a side effect of the construction, not a bolt-on.

16.3 Cooperative redeem (key path)

The happy-path redeem spends via the key path (`taproot.rs:154-179`). It is a version-2, 1-in / 1-out transaction with `nLockTime = 0` (no timelock on the happy path) and an RBF-signalling sequence. It sweeps `value − fee` to the claimer's **fresh core-wallet sweep address** — `alice_sweep_b / bob_sweep_a`, exchanged in `init/accept` (empty falls back to a key-derived address). The witness is a **single 64-byte aggregate Schnorr signature** (`SIGHASH_DEFAULT`), attached by `attach_keypath_signature` (`taproot.rs:173-179`):

```
witness = [ 64-byte aggregate Schnorr sig ]
```

Worst-case vsize: `KEYPATH_REDEEM_VSIZE = 111` (`taproot.rs:40`).

> **Warning** — The cooperative key-path redeem is **NOT RBF-bumpable**: its fee is sealed into the pre-signed adaptor sighash and cannot be re-signed unilaterally. This is mitigated by fee over-provisioning at init and a CPFP child that spends the redeem's own sweep output. See the chapter "Fees, Fee-Bumping & Auto-Refund".

16.4 Refund (script path)

The refund spends via the script path (`taproot.rs:184-234`): a single-key Schnorr signature over the CLTV tapleaf, with `nLockTime = T` (valid only once MTP ≥ T). The witness reveals the leaf and its control block:

```
witness = [ sig, refund_script, control_block ]
```

This path uses **no MuSig2 and no interactive nonce** — it is a plain BIP340 single-signature spend, signed deterministically from the refund key (`taproot.rs:219-226`). That is what makes the refund **unattended-safe**: a daemon recovering from a crash, with no live nonce state at all, can still refund from the seed alone.

Worst-case vsize: `SCRIPTPATH_REFUND_VSIZE = 140` (`taproot.rs:41`). As in v1, the engine refuses any spend whose output would fall below `DUST_LIMIT_SAT = 546` (`taproot.rs:130-133`).

16.5 The adaptor mechanism

The binding between the two legs is built before either party funds (spec v2 §6, `adaptor_engine.rs`):

1. The parties run a MuSig2 **adaptor** session over the **leg-B** redeem transaction under the point `T`, producing adaptor signature `σ_B`.
2. They run a second adaptor session over the **leg-A** redeem under the **same** `T`, producing `σ_A`.
3. **Both adaptor signatures are verified against `T` before any funding.** This is the `Signed` state in the lifecycle: reaching it guarantees that the cooperative redeems work and that Alice's broadcast will reveal `t`.

4. Alice (who knows `t`) **adapts** `σ_B` into a valid signature and broadcasts the leg-B redeem. The completed on-chain 64-byte signature now encodes `t`.
5. Bob **extracts `t`** by subtracting his adaptor contribution from the broadcast signature, adapts `σ_A`, and claims leg A before `T1`.

The symmetry to v1 is exact: instead of a preimage `s` published in a witness, the secret `t` is published *inside* the aggregate signature. Bob's daemon recovers it from chain B the same way it would scan for a v1 preimage.

16.6 Nonce safety

MuSig2 is catastrophically broken by nonce reuse — repeating a secret nonce across two different messages leaks the private key. Pact makes reuse structurally impossible (`spec v2 §3.2`):

- **Fresh from a CSPRNG.** Secret nonces are generated per BIP327 from the operating system's randomness, **never derived from the seed**. (The long-term keys and `t` are seed-derived; nonces are not.)
- **Write-ahead persisted.** A nonce is written to disk *before* it is used.
- **A monotonic state machine.** Each `(swap, leg)` nonce slot advances `none → committed → revealed → consumed` (the `nonce_sessions` store). Overwriting an already-used slot is **refused**, so the same nonce can never sign two messages.
- **Lost state falls back to refund.** If nonce state is lost (e.g. a crash before a session completes), the engine does **not** improvise a fresh nonce to finish the cooperative redeem — it lets the leg time out and takes the single-key timelock refund instead. Refund never needs a nonce, so this path is always available.

  Warning — The cardinal rule: a lost or ambiguous nonce state always resolves to the timelock refund, never to nonce reuse. Correctness is preserved at the cost of falling back from the cooperative path to the refund path.

16.7 The recovery contract

The seed and the swap index `i` always reconstruct every long-term key and the adaptor secret `t` (`keys.rs:118-128`, spec v2 §9). Therefore:

- A party can **always refund** via the single-key tapleaf using only the seed — no live session state required.
- Alice can **always re-derive `t`** to complete a redeem, even with an empty state DB.

The only thing that cannot be reconstructed from the seed is a *consumed* MuSig2 nonce — and that is by design, because reusing one would be the unsafe act. Losing nonce state costs you the cooperative path, not your funds.

16.8 Taproot tweak parity

The MuSig2 key aggregation applies the BIP341 taproot tweak via `KeyAggContext::with_taproot_tweak`, so the signing session produces a signature that verifies under the **tweaked output key**, with the parity handling that BIP340 x-only keys require. The engine's tests confirm a key-path signature attached this way verifies under the output key (`taproot.rs:341-378`), proving the tweak plumbing the aggregate signature plugs into is correct.

16.9 vsize reference

Constant	Value (vB)	Transaction
`KEYPATH_REDEEM_VSIZE`	111	cooperative key-path redeem
`SCRIPTPATH_REFUND_VSIZE`	140	single-key CLTV script-path refund
`CPFP_CHILD_VSIZE`	150	the CPFP redeem-bump child (see "Fees, Fee-Bumping & Auto-Refund")

The key-path redeem is the smaller spend precisely because its witness is a single signature with no script or control block to reveal.

Chapter 17

Timelocks & Action Deadlines

Every Pact swap is held together by two absolute Unix-time locktimes, `T1` and `T2`. They are the only thing standing between an atomic swap and a one-sided loss: if a deadline slips, the chain may let a counterparty redeem a leg the other party expected to refund. This chapter sets out the structural rule that orders the locktimes, the per-network minimums that bound their duration, and the §7.4 *action-deadline margins* that build in safety slack — and explains why those margins exist.

17.1 The structural rule: `T2 < T1`

The participant's refund locktime must always be **earlier** than the initiator's:

```
T2  <  T1
```

This is checked on every network for both protocol versions (`swap.rs:97-104` for v1, `adaptor_swap.rs:76-83` for v2). The reasoning, from the lifecycle (see the chapter "The Swap Lifecycle"):

- Alice reveals the secret when she redeems leg B; that publishes it so Bob can redeem leg A.
- Bob must have a window to redeem leg A *after* Alice reveals, but *before* Alice could refund leg A. So Alice's refund (`T1`) must come last.
- Bob's own funds on leg B must be refundable (`T2`) before that, so he is never left exposed if Alice simply walks away after funding.

If `T2 ≥ T1`, the safety ordering collapses and the swap is unsafe. The engine refuses to construct or accept such a swap.

17.2 Network-profile minimums

Beyond the ordering, mainnet and testnet enforce duration minimums so the windows are wide enough for real confirmation times and operator slack (`validate_profile`, `engine.rs:237-257`). **Regtest is exempt** so tests can use tight timelocks.

Rule	Mainnet / Testnet	Meaning
`T2 ≥ now + 3h`	yes	Bob's leg must give at least 3h before its refund
`T1 − T2 ≥ 4h`	yes	at least a 4h gap between the two refund deadlines
`T1 ≤ now + 48h`	yes	the whole swap must conclude within 48h
`N_A ≥ 6`	yes	leg A needs ≥ 6 confirmations
`N_B ≥ 1`	yes	leg B needs ≥ 1 confirmation

The `4h` gap (`T1 − T2`) is the core safety window: it is the time Bob has to redeem leg A after Alice reveals the secret, before Alice's refund could race him.

17.3 The §7.4 action-deadline margins

The profile minimums shape the *timelocks*. The §7.4 *action margins* shape the *deadlines by which the engine must act* — they pull each action earlier than its raw locktime so that a transaction broadcast late, or a daemon recovering from downtime, still has slack to confirm before the chain turns against it (`action_margins`, `engine.rs:272-278`).

Network	(fund, reveal, redeem_A)
Mainnet	(3h, 2h, 1h)
Testnet	(3h, 2h, 1h)
Regtest	(0, 0, 0)

Each margin protects a specific action:

Margin	Action it bounds	Deadline
`fund` = 3h	Bob funds leg B	no later than `T2 − 3h`
`reveal` = 2h	Alice redeems leg B (revealing the secret)	no later than `T2 − 2h`
`redeem_A` = 1h	Bob redeems leg A	before `T1 − 1h`

The engine decides whether an action is still safe to start with one formula (`engine.rs:272-278`):

```
action_safe(clock, margin, deadline)  =  clock + margin < deadline
```

If the current clock plus the margin already reaches the deadline, the action is **not** safe — the engine will not start a funding or reveal it cannot expect to finish in time, and instead steers toward refund.

17.3.1 The deadline clock

What "now" means depends on the network (`engine.rs`, deadline clock):

```
deadline_clock = max(local_time, MTP)      off regtest
deadline_clock = MTP                        on regtest
```

Off regtest, the engine takes the **later** of local wall-clock time and the chain's median-time-past (MTP) — being conservative, it never assumes the chain clock is ahead of where it actually is. On regtest it uses pure MTP so `setmocktime`-driven tests are deterministic.

17.4 Why the margins exist: the reveal-too-late race

The margins are not padding for slow blocks; they close a specific attack window — the *reveal-too-late race*. Walk it through without margins:

1. Alice waits until the very last moment before `T2` to redeem leg B and reveal the secret.
2. Her redeem confirms just barely before `T2`.
3. Bob now has the secret — but only a sliver of time before `T1` to get his leg-A redeem confirmed.
4. If Bob's redeem does not confirm in that sliver (a fee spike, a slow block, a brief outage), `T1` passes and **Alice can refund leg A** — taking back the coin Bob just paid her for. Bob has lost his leg-B coin to Alice's redeem *and* his claim on leg A. One-sided total loss.

The margins defuse this. By requiring Alice to reveal no later than `T2 − 2h` and bounding Bob's redeem to `T1 − 1h`, with `T1 − T2 ≥ 4h`, Bob is guaranteed a multi-hour window — between Alice's enforced reveal deadline and his own enforced redeem deadline — to get leg A confirmed even under adverse conditions. The engine will not let Alice's daemon reveal so late that Bob is squeezed.

> **Warning** — The margins assume both daemons are *running* through the swap. A daemon down across its reveal or redeem deadline cannot act; the margins give a recovering daemon slack, but they are not a substitute for keeping `pactd` up for the life of a funded swap.

17.5 Offer-offset validation at post time

The same constraints are checked when an offer is **posted**, on the offsets (`t2_secs`, `t1_secs`) the maker advertises, so a malformed offer never reaches the board (`validate_offer_offsets`, `engine.rs:312-330`):

```
T2 < T1
t2_secs   ≥ 3h
t1_secs − t2_secs ≥ 4h
```

Regtest is exempt here too. Validating at post time means a taker who accepts a listed offer can trust its timelocks are already sane.

17.6 Recommended values: the UI presets

Satchel's offer form ships three timelock presets, all with `T1 = 2 × T2` (`OfferForm.tsx:46-50`). These are the recommended values for real swaps:

Preset	`T2`	`T1`	Notes
Short	6h	12h	fastest turnaround; least slack
Medium (default)	12h	24h	the recommended balance
Long	18h	36h	most slack; for slow or congested conditions

All three satisfy the profile minimums (`T2 ≥ 3h`, `T1 − T2 ≥ 4h`, `T1 ≤ 48h`) with comfortable headroom. **Medium** is the default and the right choice unless you have a specific reason to widen or tighten the windows.

> **Tip** — Wider timelocks cost nothing but time-to-refund-if-things-go-wrong. If you expect fee volatility or intermittent connectivity, prefer **Long**. For a quick swap between two responsive, well-funded daemons, **Short** is fine. **Medium** is the safe default.

Chapter 18

Fees, Fee-Bumping & Auto-Refund

Atomic swaps are fee-sensitive in a way ordinary payments are not: a redeem that fails to confirm before a timelock matures can cost a party its funds (see the chapter "Timelocks & Action Deadlines"). Pact therefore treats fee-bumping and auto-refund as first-class, scheduler-driven mechanisms. This chapter covers how v1 and v2 bump stuck spends — including the important v2 asymmetry where one path can be bumped and one cannot — and how the engine fires the timelock refund automatically and safely.

18.1 v1 fee-bumping (RBF)

In v1 **both** the redeem and the refund are RBF-bumpable. Both spends signal RBF (`nSequence = 0xFFFFFFFD`), and because the v1 keys sign deterministically (ECDSA), the engine can re-sign a higher-fee replacement unilaterally (`maybe_bump`, `engine.rs:2915-2984`).

The fee escalation on each bump is roughly +50% (`engine.rs:2915-2984`):

```
new_fee = old_fee + max(old_fee / 2, MIN_SPEND_FEE_SAT)
```

with `MIN_SPEND_FEE_SAT = 1000` (`swap.rs:38`). If raising the fee further would push the swept output below `DUST_LIMIT_SAT = 546` (`swap.rs:20`), the engine stops escalating and **rebroadcasts the existing transaction** instead — a higher fee that dusts the output would be worse than simply retrying.

18.2 v2 fee-bumping: a split design

v2 is asymmetric, and the asymmetry is load-bearing (spec v2 §8):

v2 spend	Bumpable?	Why
Single-key CLTV refund	**Yes**, RBF	single-key, deterministic re-sign
Cooperative MuSig2 key-path redeem	**No**	fee sealed into the pre-signed adaptor sighash

18.2.1 The refund is RBF-bumpable

The v2 single-key refund (`adaptor_bump_refund`) bumps exactly like v1: same ~+50% escalation, deterministic single-key Schnorr re-sign, RBF sequence. No interactive ceremony is needed because the refund tapleaf is single-signature (see the chapter "v2 Taproot/MuSig2 Adaptor Swaps").

18.2.2 The cooperative redeem is NOT bumpable

The cooperative key-path redeem's fee is fixed at signing time: the fee is part of the sighash the MuSig2 adaptor session signed, and re-signing would require re-running the interactive ceremony. The engine cannot raise it after the fact. Two mitigations make this safe in practice.

(a) Over-provision the fee at init. The adaptor redeem feerate is set high *before* signing so the sealed fee is generous (`engine.rs`, `adaptor_redeem_feerate`):

```
adaptor_redeem_feerate = live_6block_estimate × ADAPTOR_REDEEM_FEERATE_MULT(3)
                         clamped to MAX_REDEEM_FEERATE = 500 sat/vB
                         fallback 20 sat/vB if no estimate
                         fixed 2 sat/vB on regtest
```

Tripling the 6-block estimate (capped at 500 sat/vB) buys headroom against fee spikes between signing and broadcast.

(b) The CPFP redeem-bump child (v2+). If the over-provisioned redeem is *still* too slow, the claimer accelerates it with a child-pays-for-parent transaction (`adaptor_cpfp_bump`, `engine.rs:1942-1985`):

- The child spends the redeem's **own vout 0** — the claimer's wallet-owned sweep output — so it is self-funded and needs no extra inputs.
- The child signals RBF, so the *child* itself can be bumped further.
- Child vsize is `CPFP_CHILD_VSIZE = 150` (`engine.rs`).
- It emits `adaptor-cpfp` / `adaptor-rebroadcast` events.

This is a **plain CPFP** (no `submitpackage` / package relay): the parent redeem stays relayable on its own, so a normal CPFP child suffices to drag it through. Proven by `test_adaptor_redeem_cpfp` (and `..._ltc`, the first v2 swap on litecoind).

> **Note** — The cooperative redeem is not RBF-bumpable, so it is handled by fee over-provisioning plus a CPFP child: enough fee is committed up front, and a CPFP child can drag the parent through if conditions tighten before the deadline. The single-key refund path is always bumpable, so the *funder* is never stuck. See the chapter "Network Support, Reorgs & Safety".

18.3 Auto-refund

The refund is the safety net: if a counterparty disappears after a leg is funded, the funder gets its coin back once the leg's timelock matures. It is scheduler-driven and clock-based — the

operator does nothing.

18.3.1 v1 auto-refund

The v1 refund is **signed and persisted at funding time** (`engine.rs:2250-2266`). The fully-signed refund transaction exists on disk the instant a leg is funded, ready to broadcast even across a daemon restart.

It fires from `try_refund_due` (`engine.rs:2877-2908`), which broadcasts only when **both** conditions hold:

```
tip_median_time_min() ≥ locktime      (the least-advanced backend's MTP has reached T)
AND the HTLC output is still unspent
```

Using the **least-advanced** backend's MTP (`tip_median_time_min`) is conservative: the engine waits until *every* watched chain agrees the timelock has matured before refunding.

Several safety details:

- **M7 guard.** `refund()` refuses to broadcast a refund that would *race a counterparty redeem*. If the counterparty has already redeemed (or could), the engine does not fire a refund that the chain would reject or that could double-spend the wrong way.
- **`-27` is success.** A node returning `-27` ("transaction already in block chain" / already known) is treated as success, not an error — the refund (or an equivalent) is already on chain.
- **Armed until N-deep.** The refund stays armed until the redeem is confirmed `N` blocks deep (spec §9.5), so a shallow reorg that un-confirms a redeem re-arms the refund.

18.3.2 v2 auto-refund

The v2 refund is **not** pre-signed at funding. It is re-derived from the seed on each call (`adaptor_refund`, `engine.rs:1599-1660`): a single-key, deterministic, unattended-safe Schnorr spend over the CLTV tapleaf. Readiness is the same MTP test:

```
tip_median_time_min() ≥ leg.locktime
```

Re-deriving from the seed (rather than persisting a signed tx) is possible because the refund key is a deterministic seed branch and the refund path needs no MuSig2 nonce — so even a daemon with an empty state DB can refund. This is the design asymmetry to remember: **v1 pre-signs the refund; v2 re-derives it.**

> **Note** — Both versions refund off the chain clock (MTP), never off local wall-clock alone. A timelock is "mature" only when the watched chains' median time past actually reaches it.

18.4 Confirmation depth as the reorg-finality knob

How many confirmations the engine waits for before treating a leg as final is the per-coin reorg-finality knob (`default_confirmations`, `engine.rs:388-394`):

Chain class	Default confirmations
Regtest	1
Fast chain (< 5-min block spacing; BTCX ≈ 120s)	10
Slow chain (≥ 5-min spacing; BTC ≈ 600s)	6

Override per coin via `Engine.coin_confirmations` (`satchel.json` → `--coin-confs`), floored at ≥ 1. Deeper confirmations mean stronger reorg protection at the cost of a slower swap; this is the dial an operator turns to trade finality against speed. See the chapter "Network Gating, Reorgs & Safety".

18.5 Fee constants reference

Constant	Value	Meaning
`MIN_SPEND_FEE_SAT`	1000 sat	floor for any HTLC/leg spend fee
`DUST_LIMIT_SAT`	546 sat	swept output must stay above this
`ADAPTOR_REDEEM_FEERATE_MULT`	3×	over-provision multiplier (v2 redeem)
`MAX_REDEEM_FEERATE`	500 sat/vB	clamp on the over-provisioned feerate
`CPFP_CHILD_VSIZE`	150 vB	the CPFP redeem-bump child

The v1 escalation step (`+max(old/2, 1000)`) and the v2 over-provision-plus-CPFP strategy are two answers to the same question — *how do I make sure a time-critical spend confirms?* — chosen because v1 can re-sign freely and v2's cooperative redeem cannot.

Chapter 19

Network Support, Reorgs & Safety

This chapter states, precisely, where Pact's swap protocols stand on safety: which versions run on which networks, how the engine picks a protocol, how confirmation depth defends against reorgs, and the safety properties that hold in every swap. Both swap protocols — v1 (HTLC) and v2 (Taproot/MuSig2 adaptor) — are reviewed, audited, and live on mainnet. The point of this chapter is to make the safety posture legible.

19.1 Network support

Both protocols run on every network — regtest, testnet, and mainnet:

Protocol	State
v1 (`pact-htlc-v1`)	Live on every network, including mainnet
v2 (`pact-htlc-v2`)	Live on every network, including mainnet

In the code, `ADAPTOR_BUILT = true` (`registry.rs:53`) and `ADAPTOR_MAINNET_ENABLED = true` (`registry.rs:59`) together make `adaptor_allowed` return `true` on every network; the test suite asserts mainnet is allowed. v1 has no network restriction.

> **Note** — Some older design docs and a few stale source comments still describe v2 as "refused on mainnet" or "not built". Those are out of date. The shipped code runs v2 on every network. Treat the registry constants (`ADAPTOR_BUILT`, `ADAPTOR_MAINNET_ENABLED`) as the source of truth.

19.2 Protocol selection prefers HTLC

When a pair could run either version, the engine **prefers v1 HTLC** (`select_protocol`). It chooses v2 adaptor only for Taproot pairs that lack an HTLC option:

- If both legs support CLTV + segwit (the v1 requirement), the engine selects **v1 HTLC**.
- v2 adaptor is selected only for taproot-capable pairs where HTLC is not available.

The shipped **BTCX ↔ BTC** pair therefore defaults to **v1 HTLC**. v2 is the more advanced construction (better privacy, single-sig refund); the engine uses the older, more-exercised path whenever it is available and v2 where it is the better or only fit.

19.3 Confirmation depth and reorg protection

A blockchain reorganization can un-confirm a transaction the engine already acted on. Pact's defense is to withhold finality until a leg is buried deep enough that a reorg is implausible (`default_confirmations`, `engine.rs:388-394`):

Chain class	Default `N`
Regtest	1
Fast chain (< 5-min spacing; BTCX ≈ 120s)	10
Slow chain (≥ 5-min spacing; BTC ≈ 600s)	6

Two behaviors flow from this:

- **Auto-redeem / completion is withheld until N-deep.** The engine does not treat a swap as done — and keeps the refund armed — until the relevant spend is confirmed `N` blocks deep. A shallow reorg that un-confirms a redeem re-arms the refund rather than leaving a party exposed.
- **Reorg alert.** If a watched HTLC or leg output *vanishes* from the chain (a reorg dropped its funding or spend), the engine raises a reorg alert so the operator and the state machine can react rather than silently proceeding on a stale view.

Override `N` per coin via `Engine.coin_confirmations` (`satchel.json` → `--coin-confs`), floored at ≥ `1`. Deeper is safer but slower.

> **Note** — Spec §7.3's suggested-minimums table lists `N_B = 3` for BTC, but the engine's heuristic `default_confirmations` returns **6** for any chain with ≥ 5-minute block spacing, including BTC. The engine default is more conservative than the spec's floor; the spec value is a *minimum*, not the shipped default.

19.4 Safety properties

Pact's crypto core — the v1 witness construction, the v2 Taproot output and tapleaf, the key derivation paths, and the adaptor reveal — matches the specs and the pinned test vectors. Two operational details are worth understanding:

- **v2's cooperative redeem is not RBF-bumpable.** Its fee is sealed into the pre-signed adaptor sighash, so it cannot be fee-bumped after the fact the way an ordinary RBF transaction can. The engine handles this two ways: fee over-provisioning at init (live 6-block estimate × 3, clamped) plus a CPFP redeem-bump child that spends the redeem's own sweep output to lift the package feerate if the network gets busy. The single-key refund

path is always bumpable, so the funder is never the stuck party. See the chapter "Fees, Fee-Bumping & Auto-Refund".

- **Relay trust is liveness-only.** Pact never trusts a relay or a chain backend for *safety* — timelocks and on-chain enforcement protect funds regardless of what any relay says. A misbehaving or offline relay can only delay a swap (a liveness failure), never steal from it. The remedy for a dead relay is the timelock refund.

  Tip — The atomicity guarantees are real and chain-enforced. As with any self-custody software, keep `pactd` running for the life of any funded swap, and prefer the **Medium** or **Long** timelock presets for larger value (see the chapter "Timelocks & Action Deadlines").

19.5 Summary

The safety model is: the chain enforces the deal, timelocks bound the worst case, confirmation depth defends against reorgs, and relays are trusted only for liveness. v2's cooperative redeem is non-bumpable by construction and is handled by fee over-provisioning plus a CPFP child. Both protocols are reviewed, audited, and live on every network.

Part V

Transports

Chapter 20

The Noticeboard Abstraction

A *noticeboard* is where offers are advertised and coordination messages are relayed. Pact treats the noticeboard as a pluggable transport behind a single trait, so the swap engine does not care whether an offer travelled over an HTTP Corkboard or a set of Nostr relays. Everything crossing the boundary is a transport-agnostic, signed `Envelope` (see the chapter "Wire Format (pact-proto)"); the relay carries only opaque, end-to-end-sealed blobs.

This chapter covers the trait itself, its two shipped implementations, and the fan-out/browse model that lets two parties trade as long as they share *one* board.

20.1 The `Noticeboard` trait

The trait lives in `libswap` at `pact/libswap/src/board.rs`. It has exactly five methods — there is no reputation, scoring, or accounts surface anywhere on it:

```
pub trait Noticeboard {
    fn post_offer(&self, offer: &Envelope) -> Result<String>;
    fn offers(&self) -> Result<Vec<Envelope>>;
    fn revoke(&self, revocation: &Envelope) -> Result<()>;
    fn relay_send_blob(&self, to: &str, blob: &str) -> Result<()>;
    fn relay_poll(&self, poll: &Envelope) -> Result<Vec<(i64, String)>>;
}
```

Method	Purpose
`post_offer`	Publish a signed `offer` envelope. Returns the offer id (the envelope `swap_id`).
`offers`	List the board's currently active offers (signed `Envelopes`).
`revoke`	Withdraw one of our own offers via a signed `revoke` envelope.

Method	Purpose
`relay_send_blob`	Store-and-forward a pre-*sealed* blob to recipient `to` (an x-only identity pubkey, hex). The blob is opaque to the board.
`relay_poll`	Fetch our coordination mail. Takes a signed `relay_poll` envelope (proving we own the recipient identity); returns `(cursor, blob)` pairs.

> **Note** — The trait is deliberately **not** `Send + Sync`. Boards are built on demand and used synchronously inside a single engine call — no `await`, no thread hop. `NostrBoard` also borrows the engine's SQLite `Store`, whose connection is `!Sync`.

20.1.1 Sealed relays, signed offers

The split between offers and relay is intentional:

- **Offers are public and signed.** Anyone may read them; the BIP340 signature proves the maker authored the terms, and listings cannot be forged.
- **Relay messages are sealed.** `relay_send_blob` only ever carries a `PACTSEALED1:` blob produced client-side. A board operator sees the recipient pubkey and ciphertext — never the sender, message type, or contents. There is no plaintext relay path. (Sealing is covered in "Wire Format (pact-proto)".)

20.2 The two implementations

20.2.1 `BoardClient` — HTTP Corkboard

`BoardClient` (`board.rs`) speaks the Corkboard HTTP API over a base URL. Each trait method maps to one HTTP call:

Method	HTTP call
`post_offer`	`POST <base>/v1/offers` → `{ "offer_id": … }`
`offers`	`GET <base>/v1/offers` → `{ "offers": [ … ] }`
`revoke`	`POST <base>/v1/offers/revoke`
`relay_send_blob`	`POST <base>/v1/relay` with `{ "to", "blob" }`
`relay_poll`	`POST <base>/v1/relay/poll` → `{ "messages": [ … ] }`

Here `cursor` is the Corkboard's own autoincrement row id. The full wire surface is documented in the chapter "Corkboard (self-hostable board)".

20.2.2 `NostrBoard` — local buffer over Nostr

`NostrBoard` (`pact/libswap/src/nostr_board.rs`) implements the same trait but performs **no network I/O at all**. Every method reads or writes the engine's `nostr_*` SQLite buffers:

- `post_offer` / `revoke` push a row onto `nostr_outbox`.
- `offers` reads the active rows of `nostr_offer_cache`.
- `relay_send_blob` queues a gift-wrap for the recipient on `nostr_outbox`.
- `relay_poll` reads `nostr_inbox` past the cursor.

A separate background service (`pactd/src/nostr_service.rs`) drains the outbox to relays and fills the inbox/cache from subscriptions. Because the inbox uses a local autoincrement id, `relay_poll` returns the *exact* `(i64, blob)` contract the HTTP board does, so the engine's cursor/dispatch loop is shared unchanged. The relay-to-event mapping is covered in "Nostr Transport (pact-nostr)".

20.3 Fan-out vs browse

The engine wires every configured board into one list, via `boards()` (`engine.rs`): one entry per comma-separated HTTP `--board-url`, plus a single aggregate entry named `"nostr"` if any Nostr relay is configured. From there, the two directions are asymmetric.

20.3.1 POST fans out to all boards

When you post or relay, the engine seals once and loops over *every* configured board. `relay_send_all` (`engine.rs`) is the canonical example:

```
fn relay_send_all(&self, to: &str, envelope: &Envelope) -> Result<()> {
    let blob = crate::board::seal_envelope(to, envelope)?;
    let mut sent = false;
    for (_, board) in self.boards()? {
        if board.relay_send_blob(to, &blob).is_ok() {
            sent = true;
        }
    }
    // … succeeds if ANY board accepted
}
```

The same one sealed blob goes to all boards; the call **succeeds if any single board accepts it**. Offer posting fans out the same way. This is best-effort redundancy: a flaky or hostile board cannot block you as long as one path works.

20.3.2 Browse selects one board

Browsing is per-board selection. `list_board_offers(sel)` (`engine.rs`, the method behind the `boardlistoffers` RPC) picks the **one** board named `sel` (an HTTP URL, or the literal `"nostr"`),

defaulting to the first configured board if `sel` is absent. It then filters out any offer you have locally revoked. The UI browses one board at a time; it does not merge listings across boards.

> **Note** — All Nostr relays collapse into a single logical board, keyed in the engine as `relay_cursor:nostr`. Whether you configure one relay or six, the browse view and the cursor treat them as one board. Adding relays adds redundancy, not separate listings.

20.4 The key property: one board in common

Because posting fans out to every board while browsing reads one, two parties need only **one board in common** to trade. The maker may post to a Corkboard *and* a half-dozen Nostr relays; the taker only has to be watching one of them. There is no global, canonical order book — just overlapping noticeboards, any one of which is enough.

20.5 The `relay_poll` cursor model

Coordination mail is pulled, not pushed. Each board exposes a **per-board monotonic cursor**:

- The engine persists the highest cursor it has processed per board (`relay_cursor:<board>`), so a poll only ever returns *new* messages.
- For the HTTP Corkboard the cursor is the `relay` table's `AUTOINCREMENT` id; for Nostr it is the local `nostr_inbox` autoincrement id. Either way, `relay_poll` returns `Vec<(i64, String)>` of `(cursor, blob)`.
- The poll envelope is **signed** (type `relay_poll`, with `body.since_id`), proving the caller controls the recipient identity, so no one can read another party's mailbox.

> **Tip** — A board reset (the SQLite file replaced) would leave a stale cursor *ahead* of every message on the fresh board. The Corkboard guards against this by serving from `0` whenever the polled cursor exceeds the maximum id it holds. See "Corkboard (self-hostable board)".

Chapter 21

Wire Format (pact-proto)

Every message that crosses a noticeboard — offers, takes, and the swap handshake — is a single, signed JSON object called an *Envelope*. The `pact-proto` crate (`pact-proto/src/`) defines the envelope, its canonical encoding, the BIP340 signing rule, the sealed-blob format used by the blind relay, and the private-offer slip codec. The crate is deliberately chain-agnostic: it knows nothing about HTLCs or Taproot. Typed message *bodies* live in the engine (`libswap`), keeping `pact-proto` a pure wire layer.

21.1 The `Envelope` struct

Defined in `pact-proto/src/envelope.rs`:

```
pub struct Envelope {
    pub v: u32,
    #[serde(rename = "type")]
    pub msg_type: String,
    pub swap_id: String,
    /// 32-byte x-only identity pubkey, hex.
    pub from: String,
    pub body: Value,
    /// 64-byte BIP340 signature, hex. Empty until signed.
    #[serde(default, skip_serializing_if = "String::is_empty")]
    pub sig: String,
}
```

Field	Wire name	Meaning
`v`	`v`	Envelope version (`u32`).
`msg_type`	`type`	Message type (serde-renamed to `type` on the wire).

Field	Wire name	Meaning
`swap_id`	`swap_id`	The swap this message belongs to.
`from`	`from`	Sender identity: a 32-byte BIP340 x-only pubkey, hex.
`body`	`body`	Opaque JSON payload; shape depends on `type`.
`sig`	`sig`	64-byte BIP340 Schnorr signature, hex. Omitted entirely when empty.

> **Note** — On the wire the field is `type`, not `msg_type`. The Rust field is renamed because `type` is a reserved word. When you craft envelopes by hand, use `"type"`.

21.2 Message types

The `type` value is one of:

`offer`, `take`, `revoke`, `relay_poll`, `init`, `accept`, `funded`, `redeemed`, `abort`

The first four are board/coordination messages; `init`/`accept`/`funded`/ `redeemed`/`abort` drive the swap handshake itself. The *bodies* for these live in the engine, not in `pact-proto`.

21.2.1 `OfferBody`

The body of an `offer` envelope (`libswap` `board.rs`):

Field	Type	Meaning
`protocol`	string	`pact-htlc-v1` or `pact-htlc-v2`.
`network`	string	`regtest` / `testnet` / `mainnet`.
`give_asset`	string	Coin id the maker gives.
`give_amount`	`u64`	Amount, smallest units.
`get_asset`	string	Coin id the maker wants.
`get_amount`	`u64`	Amount, smallest units.
`t1_secs`	`u32`	T1 timelock as a **duration**, not an absolute time.
`t2_secs`	`u32`	T2 timelock duration (`t2_secs < t1_secs`).
`ttl_secs`	`u64?`	Offer lifetime; defaults to 24h when omitted.

Field	Type	Meaning
`created`	`u64`	Unix creation time, **inside** the signed body, so expiry is verifiable from the envelope alone.

A `take` body simply echoes the maker's full signed offer back: `{ "offer": <full signed offer envelope> }`. The maker can therefore rebuild the terms statelessly and cannot be tricked into different terms — the embedded offer's signature is re-verified and must carry the maker's own identity.

21.3 Canonical JSON

Signatures commit to a *canonical* JSON encoding (`envelope.rs`, spec §8.1) so that two implementations always hash the same bytes:

- Object keys are sorted **bytewise**.
- **No whitespace** between tokens.
- Integers are rendered in **decimal**.
- **Floats are rejected** — a number that is not an `i64`/`u64` is an error, not a rounding hazard.

The canonical writer is implemented explicitly (not via serde map ordering) so that a `preserve_order` feature enabled by some unrelated dependency can never silently change a signature.

21.4 BIP340 signing

Signing (`envelope.rs`) is BIP340 Schnorr by the sender's identity key:

1. Clear `sig` (sign over the unsigned envelope).
2. Serialize to canonical JSON.
3. Compute the digest: `tagged_hash("pact/msg/v1", canonical_bytes)`.
4. Sign with deterministic `sign_schnorr_no_aux_rand`; hex-encode into `sig`.

Verification decodes `from` (32 bytes) and `sig` (64 bytes), recomputes the same digest, and checks the Schnorr signature. **The caller must additionally check that `from` matches the identity pinned for this swap** — `verify` only proves the signature is internally consistent, not that it came from the expected party.

The tagged-hash construction itself (`pact-proto/src/crypto.rs`) is the BIP340 form: `tagged_hash(tag, msg) = SHA256(SHA256(tag) || SHA256(tag) || msg)`.

21.5 `swap_id` derivation

`swap_id` is 16 hex characters (8 bytes), derived via a tagged hash (`crypto.rs`):

- v1 HTLC: `swap_id = hex(tagged_hash("pact/swapid/v1", H)[..8])`, where `H = SHA256(preimage)`.
- v2 adaptor: the same shape with tag `pact/swapid/v2` over the adaptor point `T`.

Offer swap_ids (before a concrete swap exists) are random 8-byte nonces.

21.6 The `PACTSEALED1` sealed-blob format

The blind relay never sees plaintext. A coordination envelope addressed to a recipient is sealed client-side (`pact-proto/src/seal.rs`, spec §10) into:

`PACTSEALED1:<hex(ephemeral_pubkey 33B)>:<hex(nonce 12B)>:<hex(ciphertext)>`

Construction:

1. Lift the recipient's x-only identity pubkey to its **even-Y** point.
2. Generate a fresh ephemeral secret key; ECDH against the recipient point.
3. Derive the symmetric key: `tagged_hash("pact/relay/ecdh/v1", shared)`.
4. Pick a random 12-byte nonce; encrypt the envelope JSON with **ChaCha20-Poly1305**.

Opening reverses this with the recipient's identity key (negating the secret if its parity is odd, to match the even-Y lift). The operator sees only the ephemeral pubkey, the nonce, and the ciphertext — sender, type, and contents are all hidden.

> **Warning** — There is **no plaintext downgrade**. `open_envelope` rejects any blob that does not start with `PACTSEALED1`. A board cannot inject an unencrypted envelope and have the engine parse it.

21.7 The private-offer slip codec

A *slip* is a private (off-market) offer serialized for an out-of-band channel such as a chat message (`pact-proto/src/slip.rs`). It is the **same** signed `offer` envelope, with no new fields and no protocol bump:

`pactoffer1:<base64url(canonical_json(offer_envelope))>`

The base64url is the RFC 4648 URL-safe alphabet, **unpadded** (no `=`).

`decode_slip` is the only trust gate. Before returning anything to the caller it rejects, in order:

1. an unknown or missing `pactoffer1:` prefix,
2. malformed base64url,
3. bytes that are not a valid `Envelope`,
4. an envelope whose `type` is not `"offer"`,
5. a bad BIP340 signature over the canonical JSON (verified against `from`).

Because a slip is *bearer* — anyone holding it can act on it — this validation is what guarantees a pasted slip is the maker's genuine, untampered offer before a card is ever shown. Private offers are covered end-to-end in the chapter "Private (Off-Market) Offers".

Chapter 22

Nostr Transport (pact-nostr)

Nostr is Pact's default offer/relay transport. The `pact-nostr` crate (`pact-nostr/src/lib.rs`) is **pure mapping**: it converts Pact `Envelopes` to and from Nostr events and back. It opens no relay connections, runs no async code, and touches no engine state. The relay pool, buffering, and polling live in the background service (`pactd/src/nostr_service.rs`); the `Noticeboard` facade (`NostrBoard`) lives in `libswap`. This separation is what lets the engine's existing `messages::verify` and `seal::open_envelope` still apply unchanged on the far side of a relay.

22.1 Event kinds and constants

```
pub const OFFER_KIND: u16 = 31510;   // addressable advert (NIP-33)
pub const GIFTWRAP_KIND: u16 = 1059; // sealed relay message (NIP-59)
pub const RELAY_TTL_SECS: u64 = 30 * 60;   // rolling relay TTL (1800s)
const DEFAULT_TTL_SECS: u64 = 24 * 3600;   // offer-body TTL fallback
```

22.1.1 Offers — kind `31510`

An offer maps to an **addressable** (NIP-33) event of kind `31510`:

- The `d` tag is the `swap_id`, so each `(maker pubkey, swap_id)` has exactly one current event — a re-publish replaces it.
- The `content` is the **unchanged signed offer envelope JSON**. The inner Pact signature authenticates the terms; the outer Nostr signature lets relays and clients accept the event. Both are signed by the maker's identity key.
- Plain descriptive tags carry `network`, `give`, and `get` (the give/get coin ids). Discovery subscribes by kind and filters the pair/network client-side, mirroring the HTTP board.
- The custom kind keeps non-spendable swap offers out of generic Nostr marketplace clients.

The reader (`offer_from_event`) verifies the Nostr event signature, confirms its author matches the inner `offer.from`, and returns the envelope; the engine still verifies the inner Pact signature and freshness before trusting terms.

22.1.2 Gift wraps — kind `1059`

A relay message maps to a NIP-59-style gift wrap of kind `1059`:

- The `content` is the existing `PACTSEALED1:` blob (already sealed by `pact_proto::seal`).
- A `["p", recipient]` tag addresses the recipient by x-only pubkey.
- The event is signed by a **fresh, one-time ephemeral key** (`Keys::generate()`), which hides the sender from relays.

So the seal hides the *contents* from everyone, and the ephemeral author hides the *sender* from the relay. The recipient unwraps the event and opens the seal with their identity key, which also proves the message was addressed to them.

> **Note** — One-time gift-wrap keys are never reused. A relay learns only that "pubkey X has a mailbox"; it cannot link a wrapped message back to its sender.

22.2 NIP-40 rolling expiration

Offer events carry a NIP-40 `expiration` tag, and it is **rolling**, not a fixed `ttl_secs`. For a publish at time `now`:

```
expiration = min(now + RELAY_TTL_SECS, created + ttl_secs)
```

That is, a published listing drops from relays a short while (`RELAY_TTL_SECS` = 1800s) after each publish *unless the maker re-publishes*, but never later than the offer's own final expiry (`created + ttl_secs`). The engine refreshes live offers on a shorter cadence (`Engine::REFRESH_SECS`, 10 minutes) so a genuinely live offer never lapses, while an abandoned one falls off relays quickly — a listing therefore reflects a maker who is actually online.

> **Warning** — Earlier docs described the offer expiration as `= ttl_secs`. The shipped behaviour is the rolling `min(now + 1800, created + ttl_secs)` above.

22.3 Revocation — NIP-09

Revoking an offer publishes a NIP-09 `EventDeletion` referencing the addressable coordinate `31510:<maker pubkey>:<swap_id>` via an `["a", …]` tag, telling relays to drop the listing.

22.4 Subscription filters

- **Offers:** `{ kinds: [31510] }` — subscribe by kind; pair/network filtering is client-side.
- **Mailbox:** `{ kinds: [1059], #p: [me] }` — kind-1059 events tagged to our identity.

22.5 Identity equals npub

A Pact identity *is* a Nostr npub. Both are BIP340 x-only keys over the same 32-byte secp256k1 secret, so `keys_from_secret_hex` yields a Nostr key whose public key equals the Pact identity

pubkey exactly. There is no separate Nostr account.

22.6 The sync / inbox model

`NostrBoard` does no I/O; the background service polls. Each scheduler tick runs **three steps**, so the engine lock is held only for fast SQLite work and never across a relay round-trip:

1. **`prep`** *(under the lock)* — read the active merchant's identity, pending `nostr_outbox` rows, and the offer/mailbox fetch cursors.
2. **`round`** *(lock-free)* — publish the outbox to relays and fetch new offers and gift-wrap mailbox events, with `FETCH_TIMEOUT` = 10s.
3. **`apply`** *(under the lock)* — mark outbox rows sent, insert inbox rows, upsert the offer cache, and advance the cursors.

Polling (fetch-per-tick) rather than long-lived subscriptions keeps this aligned with how the engine already drives the HTTP relay each tick and sidesteps subscription/reconnect lifecycle. Cross-relay duplicates are absorbed by `event_id` uniqueness in the buffer tables.

The buffer tables:

Table	Role
`nostr_outbox`	Offers/revocations/gift-wraps awaiting publication.
`nostr_inbox`	Received gift-wrap blobs; autoincrement id = the `relay_poll` cursor; deduped by `event_id`.
`nostr_offer_cache`	Offers discovered from relays, with expiry.

22.7 Default relay list

The default relay list lives in **Satchel**, not the engine. A fresh Satchel install is prewired with `RECOMMENDED_NOSTR_RELAYS` (`satchel/src/main.rs`), six relays:

```
wss://relay.damus.io
wss://nos.lol
wss://relay.primal.net
wss://nostr.mom
wss://nostr-pub.wellorder.net
wss://offchain.pub
```

`pactd`'s `--nostr-relay` flag defaults to **empty** — the engine itself ships no relays, so the transport is opt-in. Satchel passes its configured relays to `pactd` at launch. An explicit empty list a user saves is respected (transport off).

Tip — The Nostr transport runs *alongside* `--board-url`, not instead of it. Configure both for maximum redundancy; recall that two parties need only one board in common (see "The Noticeboard Abstraction").

Chapter 23

Corkboard (self-hostable board)

Corkboard is the HTTP noticeboard: a single `axum` + SQLite binary that anyone can run. It is deliberately dumb. It stores and serves signed offer envelopes and blind relay blobs, and does nothing else — it never matches orders, never executes a trade, never holds keys or funds, charges no fees, and has no accounts. Humans pick offers; the swap happens entirely between the two `pactd` instances. Running multiple independent operators is the goal (a Bisq-style model), which is why two parties need only one board in common (see "The Noticeboard Abstraction").

The source is `corkboard/src/main.rs`.

23.1 Running it

```
corkboard --listen 127.0.0.1:9780 --db corkboard.sqlite
```

Flag	Default	Meaning
`--listen`	`127.0.0.1:9780`	Bind address.
`--db`	`corkboard.sqlite`	SQLite database path (created if absent).

The database is created on first run. The process serves until interrupted (graceful shutdown on Ctrl-C).

23.2 HTTP API

All write endpoints require a valid BIP340 envelope signature (`pact_proto::envelope::verify`), so listings cannot be forged. The board never talks to any chain — proof-of-funds verification is the *client's* job.

Method	Path	Request	Response
GET	/health	—	"ok"
POST	/v1/offers	Envelope (type=offer)	{ "offer_id": <swap_id> }
GET	/v1/offers	query give, get, network (all optional)	{ "offers": [Envelope…] }
POST	/v1/offers/revoke	Envelope (type=revoke)	{ "revoked": <bool> }
POST	/v1/relay	{ "to": <x-only hex 32B>, "blob": <string> }	{ "id": <i64> }
POST	/v1/relay/poll	Envelope (type=relay_poll, body.since_id)	{ "messages": [{ "id": <i64>, "blob": <string> }…] }

Notes on each:

- **GET /v1/offers** returns only unrevoked, unexpired offers, newest first, LIMIT 500. The give/get/network query params filter on the offer body's give_asset/get_asset/network.
- **POST /v1/offers/revoke** flips revoked only on the signer's *own* offer (matched on offer_id **and** identity); revoked reports whether a row changed.
- **POST /v1/relay** is the **blind deposit**. It is **not** signed — by design (see Auth, below). It stores { to, blob } and returns the new row id. The board never inspects the blob.
- **POST /v1/relay/poll** is signed; it returns only the caller's own mail with id > since_id, LIMIT 100.

23.3 Storage model

Two tables (open_db):

```
CREATE TABLE offers (
    offer_id  TEXT PRIMARY KEY,
    identity  TEXT NOT NULL,
    envelope  TEXT NOT NULL,
    created   INTEGER NOT NULL,
    expires   INTEGER NOT NULL,
    revoked   INTEGER NOT NULL DEFAULT 0
);
CREATE TABLE relay (
    id        INTEGER PRIMARY KEY AUTOINCREMENT,
    recipient TEXT NOT NULL,
    blob      TEXT NOT NULL,
```

```
    created    INTEGER NOT NULL
);
```

The `relay.id` autoincrement is the cursor returned by `relay_poll`; the engine persists the highest id it has seen.

23.4 Limits and errors

Constant	Value	Effect
`MAX_BLOB_BYTES`	64 KiB	Relay blobs larger than this are rejected. The relay is for coordination envelopes, not file transfer.
`DEFAULT_OFFER_TTL_SECS`	24h	Offer TTL when the body omits `ttl_secs`.
Offer TTL cap	7 days	`ttl_secs` is clamped to a week — offers are not archives.

All errors are returned as **HTTP 400** with a body of `{ "error": <message> }`. Examples: a `to` that is not a 32-byte x-only pubkey, a blob over 64 KiB, or an envelope whose `type` does not match the endpoint.

23.5 Blind relay

The relay is content-blind. A blob arrives pre-sealed (`PACTSEALED1:` — ephemeral ECDH + ChaCha20-Poly1305, sealed in the client) and the board stores and forwards the ciphertext verbatim. The operator can see *that* recipient X has mail and how big it is, never *what* it says or *who* sent it. Sealing is covered in "Wire Format (pact-proto)".

23.6 Reset hygiene

If a board's SQLite file is wiped or replaced, a returning client's saved cursor could be *ahead* of every message on the fresh board, silently swallowing all new mail. `relay_poll` guards against this: if the caller's `since_id` exceeds the maximum id the board still holds for that recipient, it serves from `0`. Ids are `AUTOINCREMENT` (never reused), so a cursor greater than the max only happens on a real reset, not ordinary pruning. The poll is signed, so this only ever re-serves the caller's own mail.

23.7 Auth model

- **Writes to offers** (`/v1/offers`, `/v1/offers/revoke`) require a valid BIP340 envelope signature. Revocation additionally checks the offer's `identity` matches the signer.
- **`/v1/relay` deposit is unauthenticated by design.** Anyone may drop a sealed blob addressed to a pubkey — this is how a counterparty who has never met you reaches your mailbox. Confidentiality comes from the seal, not from access control.
- **`/v1/relay/poll` is signed**, proving the caller owns the recipient identity, so strangers cannot drain someone else's mailbox.
- `GET /health` and `GET /v1/offers` are unauthenticated reads.

23.8 Self-hosting walkthrough

1. **Build and run** the board:

   ```
   cargo run -p corkboard -- --listen 127.0.0.1:9780 --db corkboard.sqlite
   ```

2. **Verify** it is up:

   ```
   curl -s http://127.0.0.1:9780/health   # -> ok
   ```

3. **Point `pactd` at it** with `--board-url` (comma-separate to use several):

   ```
   pactd --data-dir ~/.pact --board-url http://127.0.0.1:9780 \
         --coin btcx=… --coin btc=…
   ```

   In Satchel, set the board URL in the noticeboard settings instead.

4. From there, posting an offer fans out to this board (and any others configured), and the two `pactd`s coordinate the swap through its blind relay.

   Tip — To expose a board beyond loopback, bind it to a routable address with `--listen` and front it with your own TLS terminator. The board itself is plain HTTP and does no transport encryption; the *payloads* it relays are already end-to-end sealed.

Chapter 24

Private (Off-Market) Offers

A *private offer* lets you trade with a specific counterparty without ever posting to a public board — "trade with a friend". The artifact is a *slip*: the exact same signed `offer` envelope that would otherwise go to the Corkboard, serialized for an out-of-band channel like a chat message. No new wire fields, no protocol bump — just a different distribution path.

24.1 The slip

A slip is the signed `offer` envelope, base64url-encoded (unpadded) with a version prefix:

`pactoffer1:<base64url(canonical_json(offer_envelope))>`

The codec lives in `pact-proto/src/slip.rs` and is described byte-for-byte in "Wire Format (pact-proto)". The decode side, `decode_slip`, is the **only trust gate**: before handing back anything it rejects a bad prefix, malformed base64url, a non-`Envelope`, an envelope whose `type` is not `"offer"`, and a bad BIP340 signature. A slip is therefore self-authenticating — a pasted slip is either the maker's genuine, untampered offer or it is refused.

24.2 Engine surface

The engine (`libswap` `engine.rs`) exposes four operations:

Operation	Behaviour
`make_private_offer(network, give, get, t1_secs, t2_secs, ttl_secs, protocol)`	Builds and signs the offer, stores it **locally** under `private_offer:<swap_id>`, and returns the slip. It does **not** post to any board.
`take_offer_slip(slip)`	Decodes and verifies the slip, runs the same gating as a board take, records a pending take, and `relay_send_all`s the sealed `take` to every configured board.
`list_private_offers()`	Lists the maker's outstanding private offers.

Operation	Behaviour
`cancel_private_offer(id)`	Marks the offer revoked and deletes its row.

Because `take_offer_slip` seals the take and fans it out over the blind relay, the taker reaches the maker even though the offer was never public — the relay addressing is by the maker's identity pubkey, which is inside the slip.

24.3 RPCs

Method	Params	Returns
`makeprivateoffer`	`give`, `get`, `t1_secs`, `t2_secs`, `protocol?`, `ttl_secs?`	`{ slip }`
`takeoffer`	`slip`	`{ taken: true }`
`listprivateoffers`	—	`{ offers }`
`cancelprivateoffer`	`offer_id`	`{ cancelled: true }`

> **Warning** — On `makeprivateoffer` the optional `protocol` is parameter **4** and `ttl_secs` is parameter **5** (positional). Earlier docs listed them in the opposite order. Use named params to avoid the ambiguity: `{ "give": "btcx:1.0", "get": "btc:0.5", "t1_secs": 86400, "t2_secs": 43200, "protocol": "pact-htlc-v1", "ttl_secs": 3600 }`.

24.4 The relay-assisted flow

1. **Maker** calls `makeprivateoffer` and gets a slip.
2. **Maker** pastes the slip into a chat (or any out-of-band channel) to the friend.
3. **Friend** calls `takeoffer` with the slip. The engine decodes/verifies it, gates the terms, and sends a sealed `take` through the blind relay of every configured board.
4. **Maker** polls their mailbox (the scheduler's `tick`), finds the take, and the swap proceeds exactly like a board-driven swap from there.

> **Note** — The maker **must be online and polling** to receive the take and drive the swap. A slip handed to a friend who acts on it while the maker is offline simply waits in the relay mailbox until the maker next polls (subject to the offer's TTL).

24.5 Bearer-slip security

A slip is a **bearer** instrument: whoever holds it can take the offer. There is no taker allow-list. Safety comes from the swap's own properties, not from secrecy of the slip:

- **Fixed terms** — the amounts, assets, and timelock durations are signed into the offer; a slip cannot be edited without breaking its signature.

- **Atomic settlement** — the swap is an atomic cross-chain swap; either both legs settle or both refund. Taking a slip cannot make the maker lose funds one-sidedly.
- **Maker-funds-first** ordering and **TTL auto-expiry** — an unused slip lapses at `created + ttl_secs`.

So a leaked slip is, at worst, an offer someone else takes on the stated terms — not a theft vector.

24.6 Fully-manual CLI fallback

Everything above is automated, but the slip is just text, so a fully manual flow works too:

```
# Maker: produce a slip and copy it out of band.
pact-cli --data-dir ~/.pact call makeprivateoffer \
  '{"give":"btcx:1.0","get":"btc:0.5","t1_secs":86400,"t2_secs":43200}'

# Friend: take it.
pact-cli --data-dir ~/.pact call takeoffer "pactoffer1:…"

# Maker: poll for the take and drive the swap.
pact-cli --data-dir ~/.pact call tick
```

For the swap handshake that follows the take, see the chapters on the v1 and v2 APIs and "The Swap Lifecycle".

> **Tip** — A slip carries **only** an `offer`. Other handshake envelopes (`take`, `init`, `accept`, …) are not slips; when moved manually they travel as raw JSON or as `PACTSEALED1:` blobs through the relay, not behind the `pactoffer1:` prefix.

Part VI

Building Against Pact

Chapter 25

Building Your Own Front-End

Satchel is the reference desktop client, but it has no privileged access: it is just one consumer of `pactd`'s JSON-RPC API. Anything Satchel does, your own front-end — a CLI, a TUI, a web dashboard, a bot — can do, by speaking the same JSON-RPC over loopback and authenticating with the cookie. This chapter is the orientation for that; per-method detail lives in the API chapters ("API: Node, Seed, Merchants, Coins", "API: v1 HTLC Swaps", "API: v2 Adaptor Swaps", "API: Board, Private Offers & Fees").

25.1 The integration contract

`pactd` is to the swap engine what `bitcoind` is to Bitcoin: a long-running daemon you drive entirely over RPC. To integrate:

1. **Speak JSON-RPC 2.0 over loopback.** `POST /` with `{ "jsonrpc": "2.0", "id", "method", "params" }`. The listen address is `127.0.0.1:9737` by default and is enforced to be loopback. Params accept a positional array or a named object.
2. **Authenticate with the cookie.** `pactd` writes `<datadir>/.cookie` holding `__cookie__:<hex>` each run; pass it as HTTP Basic credentials. A `pact.conf` with `rpcuser`/`rpcpassword` is an alternative. `GET /health` is the only unauthenticated route. See "JSON-RPC Conventions" for the full handshake.
3. **Drive the same methods** Satchel does. There is no second, private surface.

   Warning — Never expose the `pactd` port directly to a network. It binds loopback only; if you need remote access, front it with your own authenticated, encrypted tunnel.

25.2 A recommended app flow

A front-end typically walks this sequence. Each step maps to one or more RPC methods; consult the API chapters for params and return shapes.

1. **Probe the daemon.** `getinfo` and `walletstatus` — is there a seed, is it encrypted, is it locked, what coins are configured, what network.
2. **Bring up the seed.** If no seed: `createseed` (or `generateseed` → `importseed` for a preview-then-confirm flow). If encrypted and locked: `unlock` with the passphrase (or set `PACT_PASSPHRASE` at boot).
3. **Select a merchant** if running nested mode (`--merchants`): `listmerchants`, then `createmerchant` / `loadmerchant`. All swap/board calls target the active merchant.
4. **Configure and check coins.** `listcoins` (live status, tip height, genesis, capabilities), `validatecoin` before saving a backend, `listpairs` to learn which pairs and protocols are available.
5. **Browse a board.** `boardlistoffers` (a single board per call — an HTTP URL or `"nostr"`), or build your own ladder from the returned offer envelopes.
6. **Post or take.** `boardpostoffer` to advertise; `boardtake` to take an offer; or `makeprivateoffer` / `takeoffer` for off-market slips.
7. **Poll and render.** Call `tick` on an interval to advance the scheduler and drain coordination mail, then `listswaps` / `listadaptorswaps` / `listpendingtakes` / `listmyoffers` to render state.

25.3 What the front-end owns vs what `pactd` owns

The division of responsibility is strict and worth internalizing:

The front-end owns	`pactd` owns
Presentation: order-book ladder, swap cards, balances, forms.	The seed and all keys; they never leave the daemon.
Input validation for UX (amounts, pair selection).	Funding, redeeming, and **refunds** — the on-chain transactions.
When to call `tick` / how often to poll.	The scheduler that auto-funds, auto-redeems, and auto-refunds on the clock.
Choosing which board to browse.	Sealing, signing, and the noticeboard fan-out.
Storing UI preferences.	Swap state, timelock enforcement, fee bumping.

In short: your front-end is a *view and a controller*. It never holds key material and never has to implement swap safety — the engine enforces timelocks and refunds whether or not any UI is attached.

25.4 Threading and polling

- **`pactd` self-ticks.** With `--tick-secs` (default 30) the daemon runs its own scheduler loop; safety does not depend on your UI polling. A swap will refund on time even if your front-end is closed, as long as the daemon is running.

- **Poll `tick` for responsiveness.** Calling `tick` from your UI on a short interval makes state advance promptly (e.g. picking up a take, redeeming a newly confirmed leg) rather than waiting for the next self-tick. `tick` returns `{ events: [ … ] }` you can surface as activity.
- **One active merchant at a time.** RPC calls operate on the active merchant; do not assume concurrency across merchants within one process.

25.5 Error handling

Errors come back as a JSON-RPC `error` object, never an HTTP error status for business failures:

```
{ "jsonrpc": "2.0", "id": 1,
  "error": { "code": -1, "message": "no active merchant — create or load one first" } }
```

`code` is always `-1`; the human-readable cause is in `message`. Common ones to handle: `unknown method '<m>'`, `missing param '<name>'`, and `no active merchant — create or load one first` (prompt the user to create or load a merchant). Treat the presence of an `error` key — not the HTTP status — as the failure signal.

> **Tip** — For the exhaustive list of methods, their parameters, and return shapes, work from the API chapters. This chapter only sketches the flow; those are the contract.

Chapter 26

Testing & the Regtest Harness

Pact is tested at two levels: fast in-process Rust tests (unit logic plus spec-vector conformance), and a Python regtest harness that drives real `pactd` daemons and real regtest nodes through complete swaps. This chapter inventories both and shows how to run them.

26.1 The cargo test suite

`cargo test` over the workspace runs the unit tests embedded in each crate (canonical JSON, sign/verify, sealing, slip codec, the Noticeboard buffers, and the swap state machines) plus the **protocol vector tests**. The vector tests pin the wire/script construction against frozen fixtures so an accidental change to a script, key path, or hash is caught immediately:

Test file	Pins
`pact/libswap/tests/vectors.rs`	`spec/vectors/htlc_v1.json` (v1 HTLC).
`pact/libswap/tests/vectors_v2.rs`	`spec/vectors/htlc_v2.json` (v2 adaptor).

Regenerate the vectors with `cargo run -p libswap --example gen-vectors` and `cargo run -p libswap --example gen-vectors-v2` (in `pact/`); the tests then re-pin them.

```
cargo test --workspace      # unit + vector tests
cargo clippy --workspace    # lints
```

26.2 The Python harness

The harness lives in `pact/harness/`, is Python 3.10+ stdlib-only, and is built around `regtest_harness.py`:

- It brings up a **Bitcoin PoCX (btcx)** regtest node and a **Bitcoin (btc)** regtest node — always; an optional **Litecoin (ltc)** node starts only with `Harness(with_ltc=True)`.
- It funds one wallet per party per chain (e.g. `alice_pocx` funded / `bob_pocx` empty; `bob_btc` funded / `alice_btc` empty).

- It keeps both chains on a **shared mock clock** (`setmocktime`) and exposes `advance_time()` to push median-time-past (MTP) forward — this is how CLTV/timelock and refund behaviour is driven deterministically.
- Genesis hashes are asserted at startup, so a mis-pointed daemon binary fails loudly instead of silently testing the wrong chain.

The two end-to-end suites build the cargo workspace (`pactd`, `pact-cli`) and the Corkboard first, then run every scenario against a single shared `Harness`.

26.2.1 v1 HTLC scenarios — `test_swap_e2e.py`

Scenario	What it proves
`test_complete_swap`	Happy-path manual swap: two `pactd`s driven by `pact-cli` through the file handshake; Bob extracts the preimage from the chain (no courtesy message). HTLC outpoints spent on-chain, both parties received the other asset.
`test_refund`	Both fund, Alice goes silent, clocks pass T1, both refund. Negative checks: premature refunds and a late redeem past T2 are rejected (§7.4).
`test_daemon_autopilot_swap`	Alice drives everything over `pactd` JSON-RPC (cookie auth; an unauthenticated call is rejected 401); duplicated backends (multi-backend path); the scheduler auto-redeems and RBF-bumps Alice's unconfirmed redeem; books `completed` on confirmation.
`test_daemon_autopilot_refund`	Both parties walk away after funding; the `tick` scheduler alone reclaims both legs after the timelocks — and does nothing before them.
`test_chain_watched_funding`	`funded` messages are withheld; the swap still completes via on-chain funding discovery.
`test_balance_validation`	Make-offer / take balance checks reject under-funded actions.
`test_create_import_then_swap`	Seedless start: Alice `createseed`, Bob `importseed` (encrypted) via the wizard's seed-lifecycle RPCs, then a normal swap.

Scenario	What it proves
`test_coin_setup`	`listcoins` / `listpairs` / `validatecoin`: configured + connected + genesis state, capability-derived pair availability, and the genesis-hash gate (accepts the right node, rejects a cross-wired one).
`test_corkboard_swap`	Maker posts a signed offer, taker takes it, the whole handshake travels through the blind relay to completion. Covers offer withdraw, a competing second take (served-once + reject + auto-delist), and asserts every relay blob is sealed `PACTSEALED1:` ciphertext, never plaintext.
`test_board_reset_recovery`	A board reset / stale cursor is recovered (serve-from-0 hygiene).
`test_nostr_relay_swap`	A swap over a **live local Nostr relay**, offers and relay mail flowing over Nostr alone.
`test_private_offer_swap`	Off-market: `makeprivateoffer` produces a `pactoffer1:` slip posted to no board; `takeoffer <slip>` relays the take and the swap auto-completes with no board listing.

26.2.2 v2 adaptor scenarios — `test_adaptor_swap.py`

Scenario	What it proves
`test_adaptor_swap`	Happy-path v2 over RPC: the full adaptor lifecycle (`adaptorinit` → `adaptoraccept` → `adaptorfund` → `adaptornonces` → `adaptorsign` → `adaptorassemble` → `adaptorredeem`), cooperative MuSig2 key-path redeem.
`test_adaptor_refund`	Single-key CLTV-tapleaf refund (script-path, no MuSig2, unattended-safe).
`test_adaptor_refund_feebump`	The single-key refund is RBF-bumpable (deterministic re-sign).
`test_adaptor_redeem_cpfp`	CPFP child bumps an unbumpable cooperative redeem on BTC (the key-path redeem fee is sealed; a self-funded child spends the redeem's own output).
`test_adaptor_redeem_cpfp_ltc`	The same CPFP redeem-bump on litecoind — the first v2 swap on LTC.

Scenario	What it proves
`test_adaptor_depth_gate`	The reveal/redeem gate fires at the configured `--coin-confs` confirmation depth.
`test_adaptor_corkboard_swap`	A board-driven v2 swap (a PoCX↔BTC pair off-mainnet defaults to `pact-htlc-v2`, so a plain `boardpostoffer` posts a v2 offer).

26.3 Running the suites

```
cd pact/harness

# infrastructure only (nodes start, mine, mocktime works)
python regtest_harness.py --smoke

# v1 (HTLC) end-to-end suite
python test_swap_e2e.py

# v2 (Taproot/MuSig2 adaptor) end-to-end suite
python test_adaptor_swap.py
```

The harness resolves node binaries from `$POCX_BITCOIND` / `$BTC_BITCOIND` (or `bin/`), so copy your installed daemons into `pact/harness/bin/` first. PoCX regtest refuses rapid blocks unless mocktime is active, which is why the harness always sets and advances mocktime on both chains.

26.4 The playground scripts

For a full local stack you can click through (rather than assert against), use the PowerShell playground scripts in `tools/`:

Script	Brings up
`tools/playground-cork.ps1`	Regtest PoCX + BTC nodes, a **Corkboard**, two headless counterparties posting a two-sided book, and Satchel launched as a funded "Alice". Offers on PoCX↔BTC default to v2.
`tools/playground-nostr.ps1`	The same, but **relays-only**: no Corkboard, a single local ephemeral Nostr relay, Satchel with a relays-only `satchel.json`. Proves offers flow over Nostr alone.

Both block on the Satchel window and tear the whole stack down (PID/port-only) when you close it; `tools/knockdown.ps1` force-tears a stale run. Logs land in `<repo>\.playground\`.

Warning — Teardown is **PID- and port-only**, never by process name. A live mainnet `pocx-bitcoind` must not be killed by a name-based sweep; the playground nodes run on dedicated ports the teardown targets explicitly.

Chapter 27

Reference

A consolidated quick-reference for the whole handbook. Each entry points to the chapter that covers it in full.

27.1 Default ports & paths

Item	Default	Notes
`pactd` JSON-RPC	`127.0.0.1:9737`	Loopback-enforced; `POST /`, `GET /health`.
Corkboard	`127.0.0.1:9780`	`--listen`; HTTP API.
Nostr relays	none (engine)	Default 6 live in Satchel; `pactd --nostr-relay` empty.

Per-merchant data directory (flat root, or `merchants/<id>/` nested):

File	Contents
`pact.sqlite`	Swap/offer/nonce/Nostr state.
`seed.mnemonic`	BIP39 seed (plaintext, or `PACTSEEDv1:<salt>:<nonce>:<ciphertext>` when encrypted).
`.cookie`	Per-run RPC cookie `__cookie__:<hex>` (data-dir root only).
`pact.conf`	Optional `rpcuser` / `rpcpassword`.
`merchants.json`	Merchant manifest (parent data dir, nested mode).

`pact.sqlite` tables: `swaps`, `meta` (counters / `relay_cursor` / private offers), `pending_takes`, `nonce_sessions`, `adaptor_swaps`, `nostr_outbox`, `nostr_inbox`, `nostr_offer_cache`, `my_offers`.

Corkboard SQLite tables: `offers`, `relay`. See "Corkboard (self-hostable board)".

27.2 RPC method index

All methods are JSON-RPC over `POST /`. Grouped by area; see the named chapter for params and return shapes.

Node / info (*"API: Node, Seed, Merchants, Coins"*): `getinfo`, `walletstatus`, `stop`.

Seed lifecycle (same chapter): `createseed`, `generateseed`, `importseed`, `unlock`.

Merchants (same chapter): `createmerchant`, `listmerchants`, `loadmerchant`, `unloadmerchant`, `getmerchantinfo`.

Coins / pairs (same chapter): `listcoins`, `listpairs`, `validatecoin`.

Wallet helpers (same chapter): `getbalance`, `getnewaddress`, `sendtoaddress`.

Swaps — v1 HTLC (*"API: v1 HTLC Swaps"*): `listswaps`, `getswap`, `listpendingtakes`, `listmyoffers`, `offer`, `acceptoffer`, `recv`, `fund`, `redeem`, `refund`, `abort`, `tick`.

Swaps — v2 adaptor (*"API: v2 Adaptor Swaps"*): `listadaptorswaps`, `adaptorinit`, `adaptoraccept`, `adaptorrecv`, `adaptorfundingready`, `adaptornonces`, `adaptorsign`, `adaptorassemble`, `adaptorfund`, `adaptorredeem`, `adaptorrefund`.

Board (*"API: Board, Private Offers & Fees"*): `boardlistoffers`, `boardstatus`, `boardpostoffer`, `boardtake`, `boardrevoke`.

Private offers (same chapter, and "Private (Off-Market) Offers"): `makeprivateoffer`, `takeoffer`, `listprivateoffers`, `cancelprivateoffer`.

Fees (same chapter): `estimateswapfees`.

27.3 Error format

Failures return a JSON-RPC `error` object (not an HTTP error status):

```
{ "jsonrpc": "2.0", "id": 1, "error": { "code": -1, "message": "<reason>" } }
```

`code` is always `-1`. Common messages: `unknown method '<m>'`, `missing param '<name>'`, `no active merchant — create or load one first`. See "JSON-RPC Conventions".

27.4 BIP32 derivation paths

All keys derive from one BIP39 seed under purpose `7228'`. `coin(c)`: BTC = `0`, PoCX = `0x504F4358`. See "HTLC v1 (Construction)" and "v2 Adaptor (Construction)".

Key	Path	Use
Identity	`m/7228'/0'/0'`	BIP340 x-only; signs envelopes; equals the Nostr npub. Never used in an HTLC.
Swap key	`m/7228'/1'/coin(c)'/i'`	One compressed secp key per chain per swap (ECDSA in v1; reused as the MuSig2 x-only signer in v2).
Preimage source	`m/7228'/2'/i'`	`s = Tagged-Hash("pact/htlc/preimage/v1", priv)`, `H = SHA256(s)` (v1, Alice-only).
Adaptor secret source	`m/7228'/2'/i'`	`t = Tagged-Hash("pact/adaptor/secret/v2", priv) mod n`, `T = t·G` (v2, Alice-only).
Refund key (v2)	`m/7228'/3'/coin(c)'/i'`	x-only single-key CLTV tapleaf, independent of MuSig2.

27.5 Tagged-hash tags

`tagged_hash(tag, msg) = SHA256(SHA256(tag) || SHA256(tag) || msg)`.

Tag	Used for
`pact/msg/v1`	Envelope signing digest.
`pact/swapid/v1`	v1 `swap_id` (over `H`).
`pact/swapid/v2`	v2 `swap_id` (over `T`).
`pact/htlc/preimage/v1`	v1 preimage `s`.
`pact/adaptor/secret/v2`	v2 adaptor secret `t`.
`pact/relay/ecdh/v1`	Sealed-blob symmetric key (ECDH → ChaCha20-Poly1305).

27.6 Spec & vectors file map

File	Contents
`spec/protocol.md`	HTLC v1 (`pact-htlc-v1`): scripts, tx templates, key paths, preimage/timelock rules, the §8 message handshake.

File	Contents
`spec/protocol-v2.md`	v2 (`pact-htlc-v2`): Taproot/MuSig2 adaptor swaps; specifies only what changes from v1.
`spec/vectors/htlc_v1.json`	v1 test vectors (pinned by `tests/vectors.rs`).
`spec/vectors/htlc_v2.json`	v2 test vectors (pinned by `tests/vectors_v2.rs`).

27.7 Glossary

- **HTLC** — Hash Time-Locked Contract; the v1 swap output: spendable by a hash preimage (redeem) or after a CLTV timelock (refund).
- **Adaptor signature** — a signature missing a secret scalar; completing it on one chain reveals that scalar, letting the counterparty complete the other. The basis of v2 swaps.
- **MuSig2** — a two-round Schnorr multisignature; the 2-of-2 aggregate that forms the v2 Taproot internal key.
- **Tapleaf** — a script leaf in a Taproot tree; v2 uses one, the single-key CLTV refund script.
- **CLTV** — `OP_CHECKLOCKTIMEVERIFY`; enforces an absolute locktime, used for the refund branch. Pact requires Unix-time locktimes (`≥ 500000000`).
- **MTP** — median time past; the 11-block median a node uses to evaluate time-based locktimes. Refund readiness is keyed on MTP.
- **RBF** — Replace-By-Fee; rebroadcasting a tx with a higher fee. v1 redeem and refund and the v2 single-key refund are RBF-bumpable.
- **CPFP** — Child-Pays-For-Parent; a child tx that bumps a stuck parent's effective fee. Used to accelerate the **unbumpable** v2 cooperative redeem.
- **Preimage** — the secret `s` whose hash `H = SHA256(s)` locks a v1 HTLC; revealing it on one chain unlocks the other.
- **Sweep address** — a fresh wallet-owned address a v2 cooperative redeem sweeps to (`alice_sweep_b` / `bob_sweep_a`).
- **Merchant** — one seed bound to one data directory; the engine's wallet analog. RPC calls target the active merchant.
- **Noticeboard** — the pluggable offer/relay transport behind the five-method `Noticeboard` trait (Corkboard over HTTP, or Nostr).
- **Slip** — a private (off-market) offer serialized as `pactoffer1:<base64url>`; the same signed `offer` envelope, never posted to a board.